राष्ट्रीय डिज़ाइन संस्थान
NATIONAL INSTITUTE OF DESIGN
GANDHINAGAR

Collaborative by Design

Supporting Diverse Learning Expressions at NID 2030

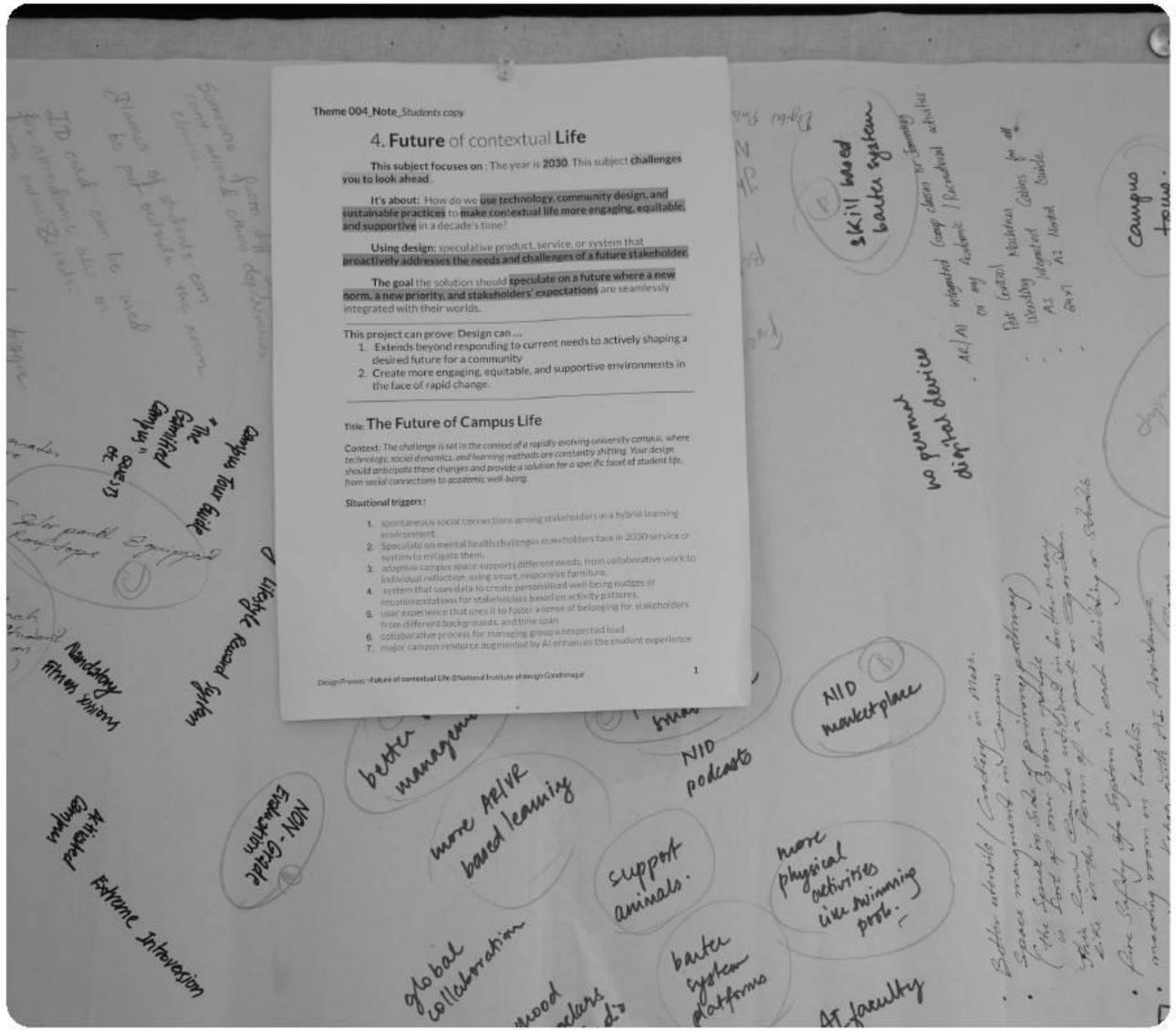
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Part I

Discovery and Definition

1.0 Introduction & Theme.
2.0 Research & Empathy
3.0 Synthesis & Insights
4.0 Design Brief & Problem Framing



1.0 Introduction & Theme

1. **Project Title:** The Future of Campus Life
   Project Theme: Future of Contextual Life

Project Context

The challenge is set in the context of a rapidly evolving university campus, where technology, social dynamics, and learning methods are constantly shifting. The design should anticipate these changes and provide a solution for a specific facet of student life, from social connections to academic well-being.





2. Context

Our intervention exists within NID's evolving hybrid learning ecosystem of 2030, a space where collaboration unfolds fluidly between physical studios and digital platforms.

Learning no longer happens in isolated classrooms but across connected environments that adapt to diverse expressive and engagement styles.
The context of our project spans:

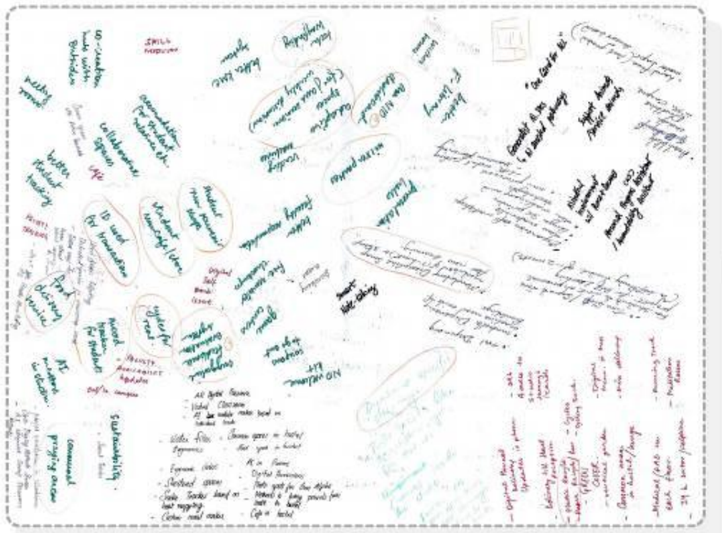
- Physical studios – for sensory, spatial, and social interactions.
- Digital environments – for asynchronous, personalized, and inclusive participation.

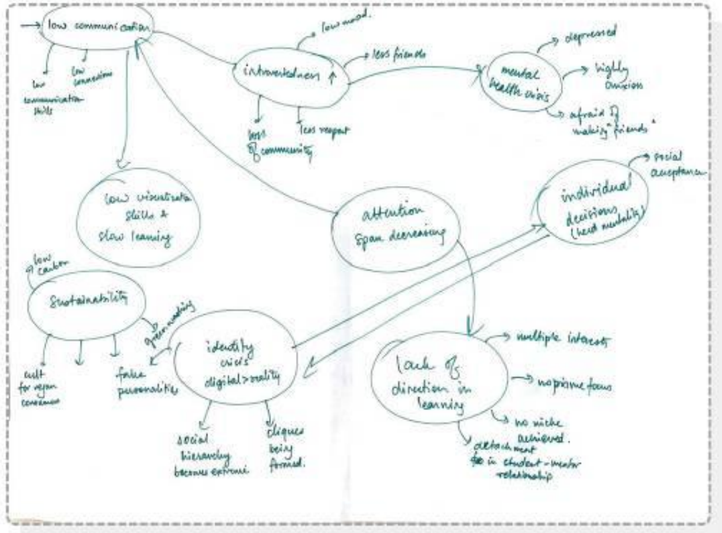
Our design interventions aim to enable every learner, regardless of expression style, to find their rhythm, connect, and contribute meaningfully.

2.0 Research & Empathy

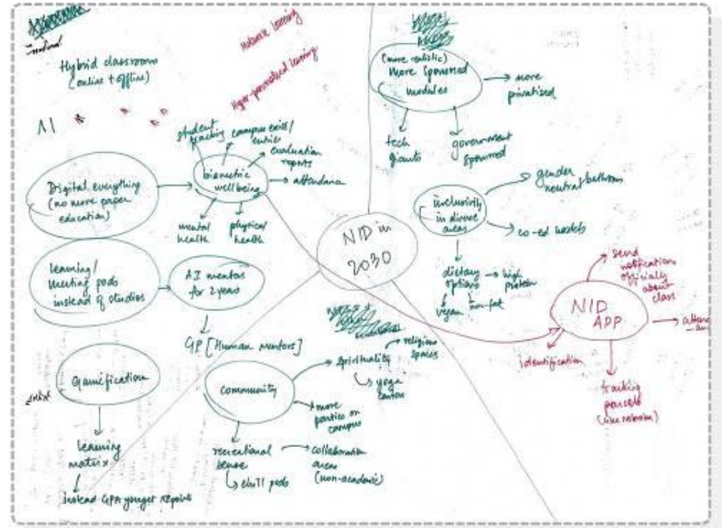
1. Defined Problem

Students at NID exhibit diverse expressive and engagement styles. Some thrive in verbal, visible collaboration, while others process ideas internally through making or reflection. Current systems tend to favor more outspoken learners, leading to uneven participation and collaboration experiences.









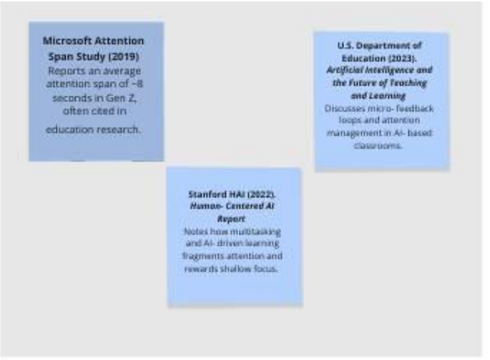
Part I: Discovery and Definition

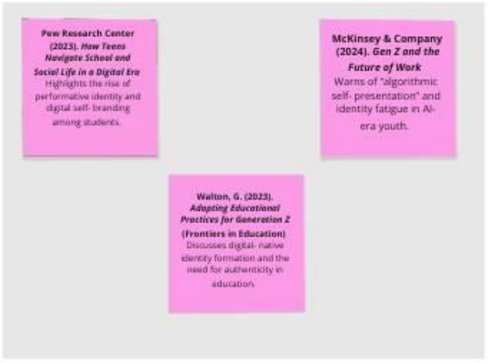
Secondary Research: Literature Study

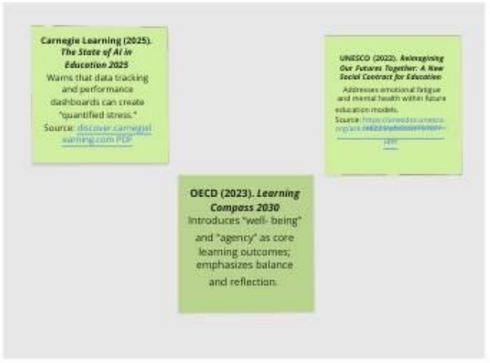
Our initial assumptions about the future of learning were grounded in lived experience and speculative brainstorming. To validate and expand on these intuitions, we conducted a targeted literature review of global reports, educational research, and foresight documents addressing learning behavior, attention, digital identity, and educational transformation.

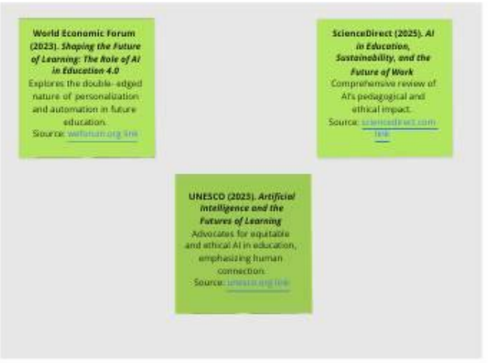












Studies revealed recurring patterns:

- Bias toward expressive learners and limited accommodation for reflective participation.
- Attention fragmentation and quantified stress due to AI-based learning tools and constant feedback loops.
- Rise of introversion, identity fatigue, and digital loneliness in an increasingly mediated campus culture.
- Need for reflective, low-stimulation environments to support diverse learning energies.
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These insights helped us validate our assumptions and build a futures-oriented framework, TESEEC, to map the key forces shaping design education at NID in 2030.

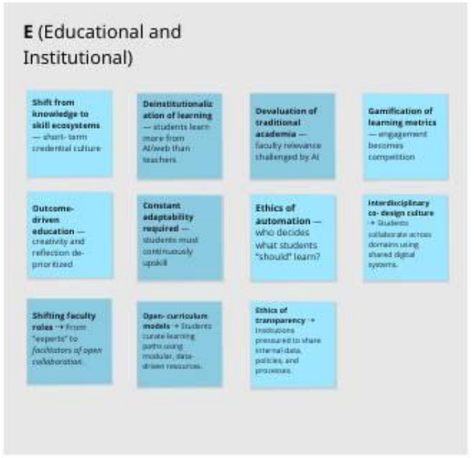
TESEEC Framework

The TESEEC framework was developed to organize and synthesize the major forces affecting the future of learning. Each driver represents a cluster of insights drawn from both literature and speculative analysis.

- Technological: AI-driven personalization, datafication, and emotional surveillance redefining how students learn and are assessed.
- Educational: Persistent "quiet bias" and the need for inclusive, reflective learning models that value varied expressions of engagement.
- Social: Shifting peer dynamics, digital isolation, and the erosion of spontaneous community among students.
- Environmental: Cognitive overload and overstimulation, calling for hybrid and sensory-balanced studio environments.
- Economic: Platformization and privatization of learning, deepening inequalities in educational access and autonomy.
- Cultural: Changing norms of expression and identity, revaluing introspection, and authenticity as valid modes of participation.

Together, these six forces shaped the speculative scenarios and interventions envisioned for NID 2030.













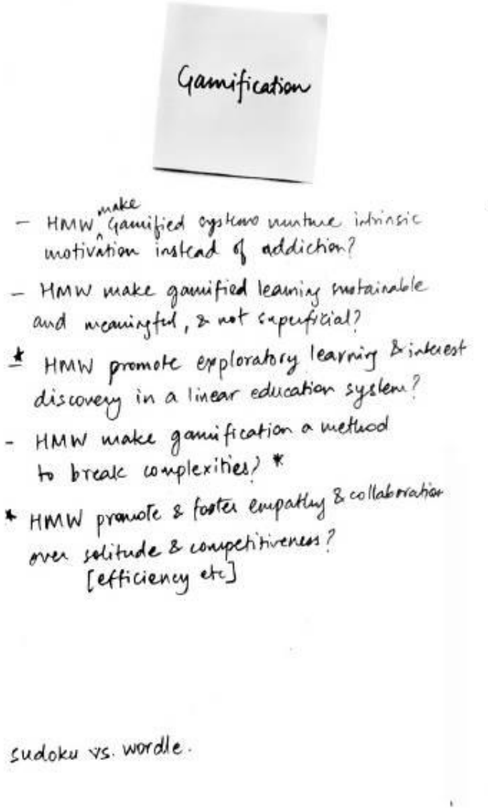
HMW framework applied to give us an idea of what has more possibilities for design intervention

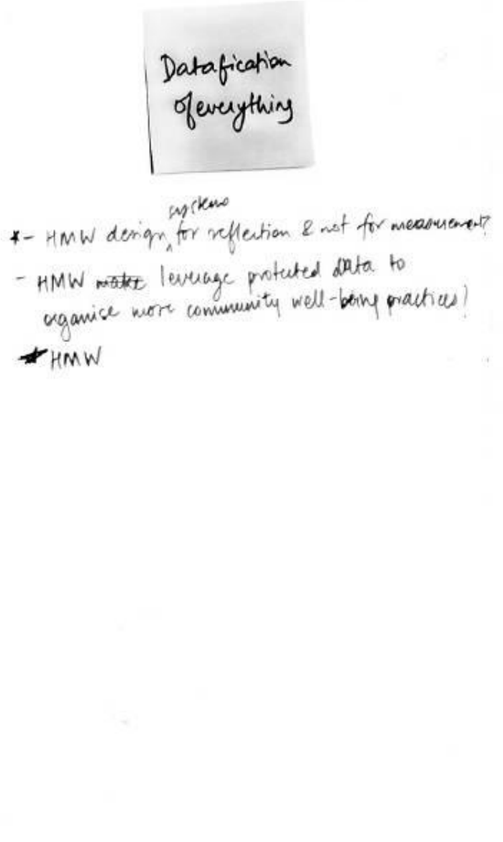
We scored each of them on the basis of:

1. Relevance: Does this respond to current/future needs of campus life? Also make ensure its meaningful for learning.
2. Human Depth: Does this involve people's emotions, habits, or behaviours, not just systems? This would ensure empathy.
3. Design Opportunity: Can design actually intervene here (visual, interaction, spatial, systemic)? to ensure feasibility
4. Scalability: Can this idea grow from NID to broader society?

And finally, our own intuition on what problem made most sense to address.







2. Methodologies

We used a combination of stakeholder mapping, semantic differential scaling, directed storytelling interviews, and cultural probes to understand different learner archetypes and their collaboration challenges.
Followed by empathy mapping and affinity mapping to identify recurring patterns of energy, expression, and participation.

Why these methods:

- Stakeholder mapping → To identify key actors shaping learning culture.
- Semantic differential scaling → To quantify expression and engagement on a spectrum.
- Directed storytelling & probes → To elicit emotional, personal narratives from participants.
- Empathy & affinity mapping → To synthesize qualitative and quantitative insights into clear behavioral clusters.

Stakeholder Mapping

1. Core (Direct) Stakeholders

These groups are essential to daily operations and are the direct focus of the institution's activities.

- Students
- Faculty Mentors
- NID Admin and Management
- Technicians Workshop Staff
- Admin Academic Coordinators
- Hostel/Mess Staff

2. Institutional (Indirect) Stakeholders

These groups provide formal support, resources, or oversight.

- NID Alumni Network
- Parents/Families
- Accrediting Bodies (Design Councils, etc.)
- Research Labs/Think Tanks
- Tech Partners/Software Providers
- Placement/Infrastructure Systems

3. External & Contextual Stakeholders

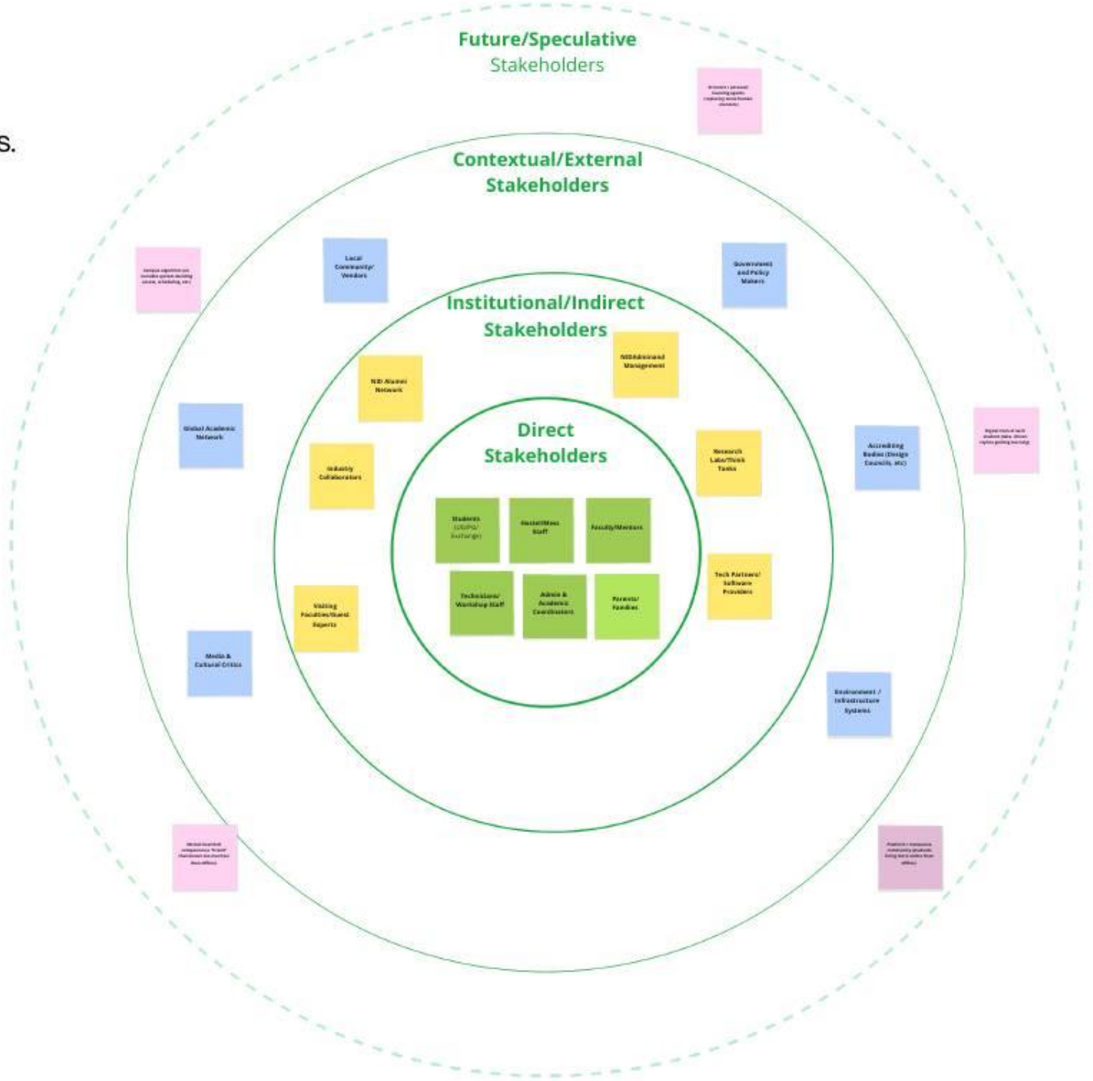
These entities shape the environment, provide external validation, or form partnerships.

- Government and Policy Makers
- Industry Collaborators
- Global Academic Network & Visiting Faculties/Guest Experts
- Media & Cultural Critics
- Local Community Vendors

4. Future & Speculative Stakeholders

- Future/Speculative Stakeholders

Focus: Considering the long-term impact and future relevance of our decisions.



Semantic Differential Scale - Framing Expression as a Learning Spectrum: Basis for the Survey

Axis & Core Focus	Reflective Attributes	Expressive Attributes	Why it Matters
Energy Orientation	Gains energy from solitude/calm. May need recovery after critique.	Gains energy from interaction/exchange.Recharged after team sessions.	Affects engagement rhythm, burnout, and social fatigue.
Processing Mode	Thinks before speaking.Prefers private exploration.	Thinks by speaking. Clarifies through quick feedback.	Defines contribution to discussions and handling ambiguity.
Preferred Environment	Quiet, contained. Works best alone/small groups.	Stimulating, active. Works best in open studios/collaboration.	Helps understand spatial design needs for inclusivity.
Feedback Style	Likes written, structured, or delayed critique.Reflects privately.	Likes verbal, spontaneous, in-the-moment.Adapts on the go.	Defines emotional comfort in critique culture.
Expression Mode	Expresses best through visuals, writing, or making. Large-group articulation is stressful.	Expresses best through speaking and presenting. Enjoys visibility moments.	Ties to confidence, presentation, and classroom equity.
Participation Style	Through listening/notes/subtle contributions. Learning is inward}.	Through conversation/initiative/visible leadership. Learning is outward}.	Gets at the "visibility bias" in design education.
Stimulation Threshold	Overstimulated easily. Prefers deep focus zones/withdrawal.	Energized by movement/buzz. Needs variety and activity.	Helps explain attention, studio layout, and burnout patterns.

Semantic Differential Scale - Framing Expression as a Learning Spectrum

Instructions:

- For each statement, mark a point between the two ends that best represents you.
- There are no right or wrong answers — this is just to understand how different learners engage.

Scale (Reflective<>Expressive):

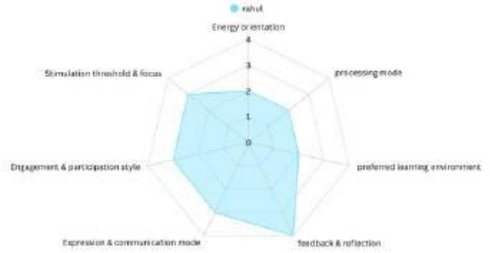
- Strongly towards left anchor
- Slightly towards left
- Leaning left
- Leaning right
- Slightly towards right
- Strongly towards right

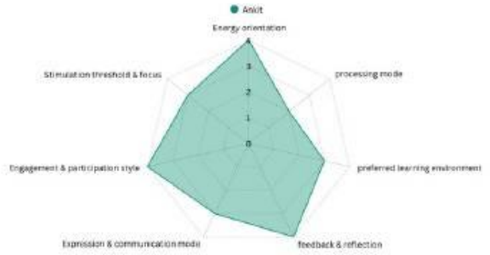
Axis & Core Focus	QUESTION 01	QUESTION 02
Energy Orientation How you recharge and sustain motivation	After a full day of studio work, I usually feel → Drained ☐ ☐ ☐ ☐ ☐ ☐ Energized	Working in a group helps me regain energy → False ☐ ☐ ☐ ☐ ☐ ☐ True
Processing Mode How you form and clarify ideas	I understand my thoughts best when I → Think or sketch quietly ☐ ☐ ☐ ☐ ☐ ☐ Talk them through aloud	When given a new brief, I prefer to → Reflect first in private ☐ ☐ ☐ ☐ ☐ ☐ Discuss ideas immediately
Learning Environment The sensory & spatial setup you thrive in	I do my best work in spaces that are → Quiet and contained ☐ ☐ ☐ ☐ ☐ ☐ Lively and social	Being surrounded by classmates while working feels → Distracting ☐ ☐ ☐ ☐ ☐ ☐ Motivating
Feedback & Reflection Style How you prefer critique and evaluation	I prefer feedback that is → Written / structured ☐ ☐ ☐ ☐ ☐ ☐ Verbal / spontaneous	I process feedback best when → I take time alone to reflect ☐ ☐ ☐ ☐ ☐ ☐ I talk it through with others
Expression & Communication Mode How you express ideas and confidence	Presenting my work in front of others feels → Stressful ☐ ☐ ☐ ☐ ☐ ☐ Exciting	I express my ideas best through → Making or writing ☐ ☐ ☐ ☐ ☐ ☐ Speaking or performing
Engagement & Participation Style How visible your learning seems	In discussions, I contribute by → Listening and absorbing ☐ ☐ ☐ ☐ ☐ ☐ Speaking and debating	When my ideas are challenged, I → Retreat to rethink quietly ☐ ☐ ☐ ☐ ☐ ☐ Engage and defend them
Stimulation Threshold How you handle input and sensory load	When there's a lot happening in the studio, I → Lose focus ☐ ☐ ☐ ☐ ☐ ☐ Feel	I stay most focused when → The space is calm and predictable ☐ ☐ ☐ ☐ ☐ ☐ There's energy and activity around

Semantic Differential Scale - Framing Expression as a Learning Spectrum: *Learnings*

7 types of archetype in NID

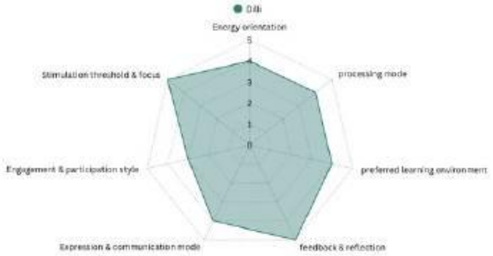
The Quiet Contributor:

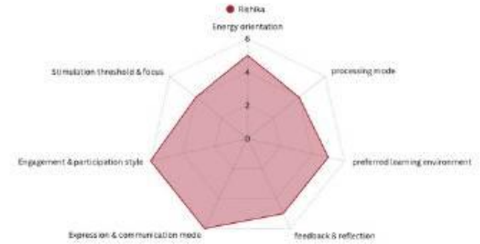




These subjects have a very low score on Stimulation Threshold, meaning they are easily distracted in a chaotic, open-plan studio. This can occur regardless of their other scores.

The Social Butterfly

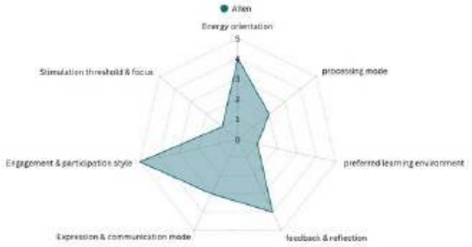


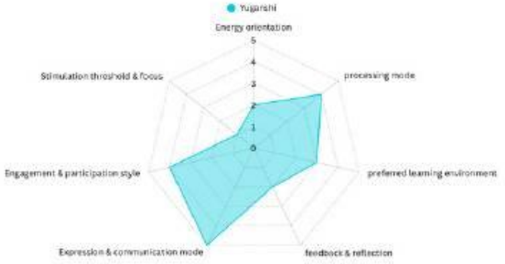


These subjects have consistently high scores across all axes, indicating a clear preference for lively, expressive, and interactive learning environments.

The Ambiguous Ambivert

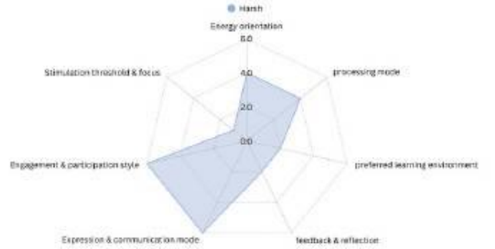
(the most popular)

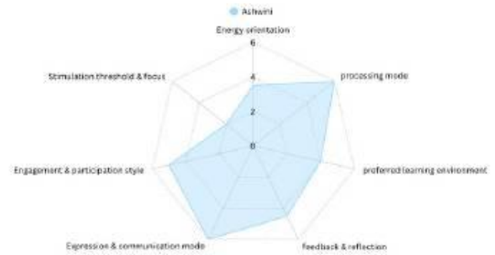




These subjects identify as ambiverts but have scores that don't neatly fit in the middle. Their data shows a subtle lean toward either the reflective or expressive end of the spectrum.

The Adaptive Performer



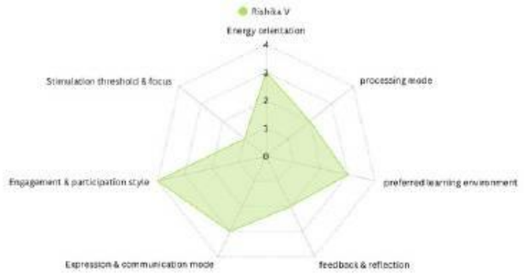


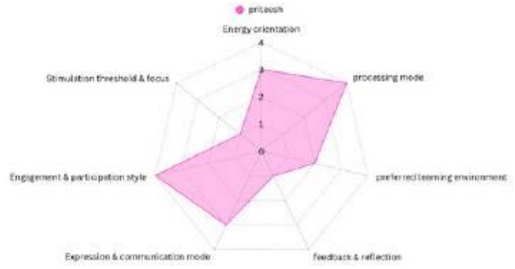
These subjects have a low-to-mid score on Energy Orientation, meaning they get drained by social interaction, but a high score on Expression and Participation, indicating they perform well in public settings.

Semantic Differential Scale - Framing Expression as a Learning Spectrum: *Learnings*

7 types of archetype in NID

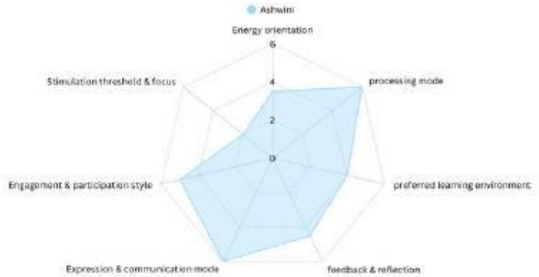
The Overwhelmed Learner

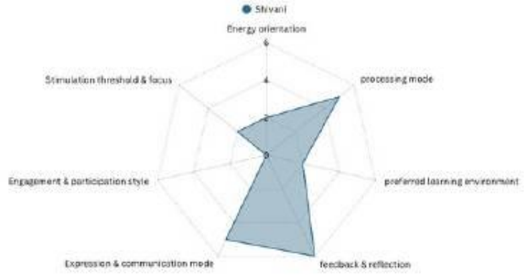




These subjects have a very low score on Stimulation Threshold, meaning they are easily distracted in a chaotic, open-plan studio. This can occur regardless of their other scores.

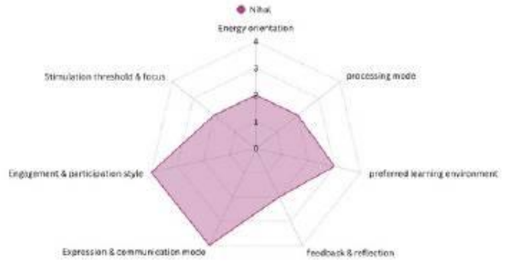
The Mismatched Receiver

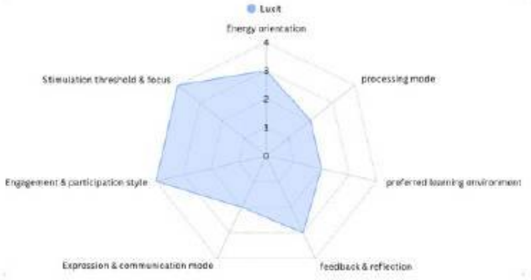




These subjects learn by talking and discussing ideas but prefer to receive critique in a private, structured format.

The Consistent Introvert





These subjects identify as introverts but have a powerful contradiction between their self-perception and their behavioral scores. Their data shows they exhibit expressive behaviors, creating a powerful disconnect.

Directed Storytelling Methodology

Understanding How Students Learn, Belong, and Thrive at NID

We designed this interview to explore how different kinds of students experience learning and collaboration at NID.

Rather than focusing on performance, we aimed to understand how personality, space, and systems interact to shape a student's experience of design education.

Our goal:
To uncover insights about belonging, learning preferences, collaboration patterns, and how NID's systems can evolve to support diverse learners by 2030.

Theme	What It Explores	Why We Asked It
1. Coming to NID: "Adjusting, Discovering, Belonging"	First encounters, early coping, social initiation	To understand how new students find belonging or isolation when entering a high-collaboration environment.
2. Learning & Creative Flow: "How You Learn Best"	Personal learning rhythm, autonomy, reflection, sensory needs	To identify intrinsic learning patterns and what environments enable or inhibit focus.
3. Collaboration & Classroom Stories: "Working With Others"	Team dynamics, conflict, validation, emotional labor	To see how group structures affect motivation, participation, and confidence.
4. Design Education Fit: "How the System Feels"	Alignment with NID's teaching culture	To locate frictions between personal learning style and institutional methods.
5. Speculative Aspirations: "If NID Were Reimagined for You"	Future-facing imagination, role of tech, ideal systems	To encourage participants to imagine how design education could evolve by 2030.
6. Closure: "Bringing It Back to You"	Reflection, unmet needs, emotional resonance	To understand what students wish their peers and faculty recognized about their ways of learning.

Directed Storytelling Methodology

Understanding How Students Learn, Belong, and Thrive at NID

Why Storytelling Over Questioning?

Using Directed Storytelling Instead of Direct Questions

Instead of a standard Q&A, we used the directed storytelling method, encouraging students to narrate personal moments rather than give quick answers.

This helps surface richer emotional cues, hidden motivations, and tacit knowledge, things that structured surveys can't capture.

We listened for:

- **What they say** → explicit thoughts, judgments, quotes
- **What they do** → behavioral cues, coping patterns
- **What they feel** → emotions, triggers, moments of comfort/discomfort
- **What they need** → underlying desires or systemic gaps

1. COMING TO NID: "Adjusting, Discovering, Belonging"

Main Prompt	Follow-ups / Probes	Listen For
Was there a moment/activity you felt you truly fit in? Or a moment you felt out of place?		Belonging, alienation, early coping patterns
How do you usually form new friendships or teams here?	Is that easy or hard for you?	Social initiation styles, energy balance

2. LEARNING & CREATIVE FLOW : "How You Learn Best"

Main Prompt	Follow-ups / Probes	Listen For
Think about a recent project or module that went really well for you.	When did you know you were doing well?	Flow moments, autonomy, learning patterns
Now tell me about one that didn't go as smoothly.	What made it challenging? How did you handle that?	Triggers for stress, self-regulation
If you could design your ideal workspace or studio environment, what would it be (look and sound) like?	Why those elements?	Sensory and spatial needs, environmental triggers

3. COLLABORATION & CLASSROOM STORIES : "Working With Others"

Main Prompt	Follow-ups / Probes	Listen For
Tell me about a team project that felt really good to be part of.	What made the team click? What role did you take naturally?	Collaboration strengths, group energy
Now think of a project that frustrated or drained you.	What happened? How did you respond?	Conflict, group pressure, emotional fatigue
What's usually going through your head during critiques or juries?	What parts do you enjoy / dread? What do you do after a jury?	Anxiety, self-presentation, validation dynamics

4. DESIGN EDUCATION FIT : "How the System Feels"

Main Prompt	Follow-ups / Probes	Listen For
When you think of NID's teaching style : studios, juries, collaboration : what fits you best? What doesn't? What would you change about it?	Why do you think that is? How would it make the learning experience better?	Alignment / misalignment with system

5. SPECULATIVE ASPIRATIONS : "If NID Were Reimagined for You"

Main Prompt	Follow-ups / Probes	Listen For
What would an ideal learning environment for someone like you look like if there were no physical limitations or realism involved?	How would you want to interact, collaborate, or recharge?	Desired systems, adaptive environments
What role do you think technology or AI could play in that version of NID?	Why?	TESEEC drivers, speculative anxieties
What should NID of the future keep exactly the same?	Why is that important to you?	Core values, identity of institution

6. CLOSURE : "Bringing It Back to You"

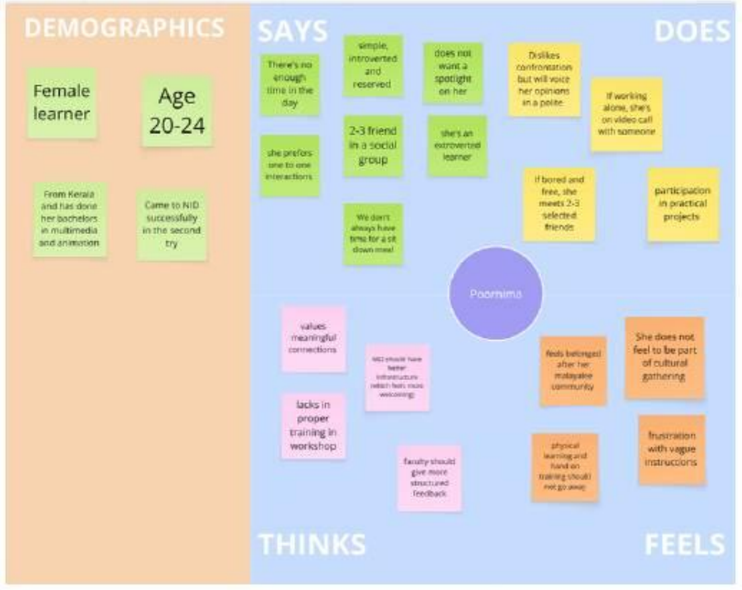
Main Prompt	Follow-ups / Probes	Listen For
Is there something about how you learn or collaborate that you wish your peers or faculty understood better?	What would change if they did?	Self-awareness, emotional needs

3.0 Synthesis & Insights

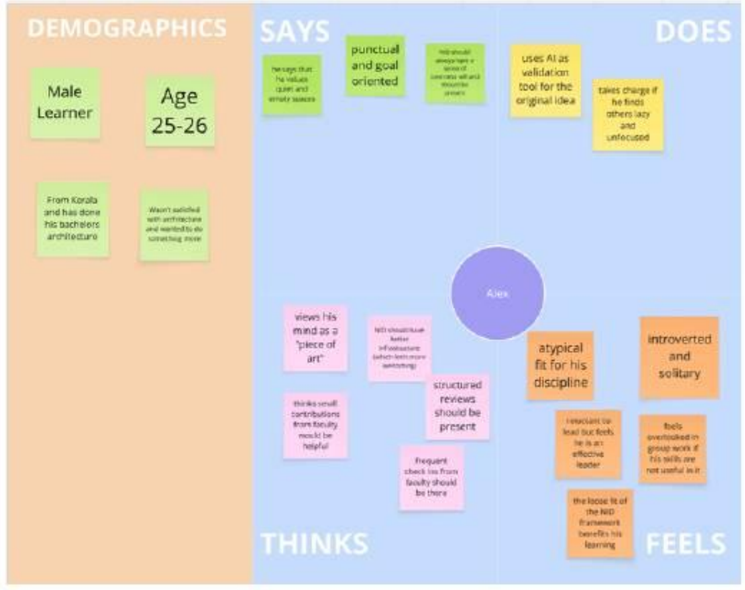
Empathy Mapping based on Directed Storytelling

- Overwhelm is a Universal Barrier: All subjects express feeling overwhelmed either by the sheer volume of information (Subject 1 & 2) or the difficulty of efficiently integrating learning with life/work (Subject 3).
- Community and Social Validation are Critical: There is a consistent desire across all demographics for peer support, mentorship (buddies, support groups), and validation/feedback on their work. The learning journey is not just individual but social.
- Need for Structure and Starting Points: Learners need a "blueprint" (Subject 2) or a simple starting point (Subject 1) to overcome analysis paralysis and procrastination. There is a clear demand for structured content and practical projects with clear objectives.
- Beyond Technical Skills: Subject 1 highlights a specific need for training in soft skills, suggesting that a holistic approach to professional development is valued alongside technical knowledge.

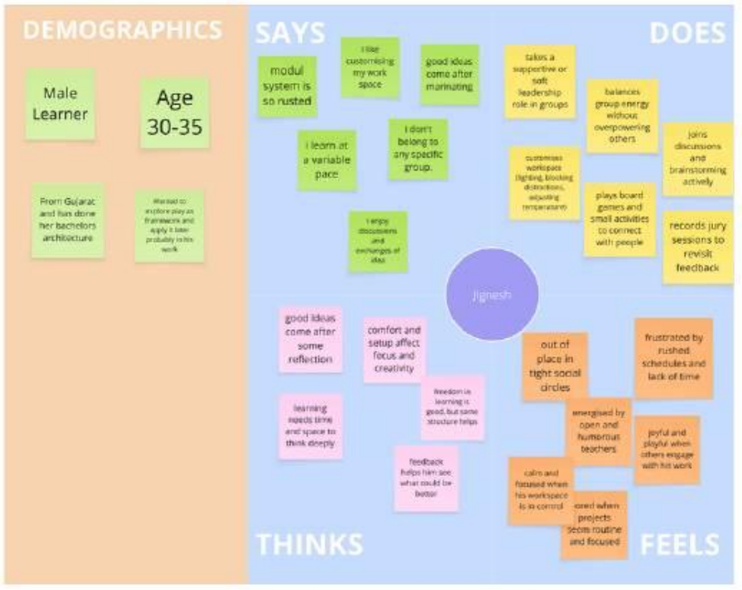
Subject 1



Subject 2



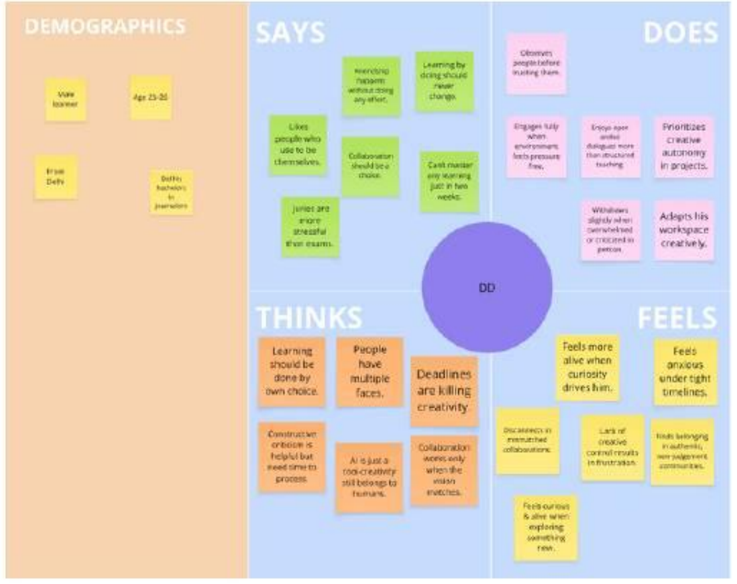
Subject 3



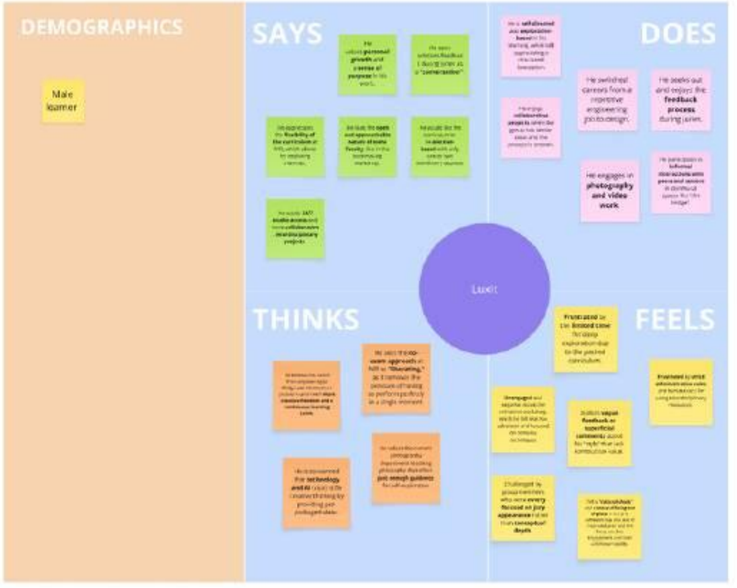
Empathy Mapping based on Directed Storytelling

- The "Creative Block" is a Major Barrier: All three subjects are hindered by the gap between inspiration/creativity and execution. They need a system that facilitates starting, maintains motivation, and encourages the completion of projects.
- Need for Validation and Feedback: Validation is crucial for this group, whether it's seeking feedback on designs (Subject 6), connecting with others to share ideas (Subject 4), or overcoming self-doubt (Subject 5 & 6). The learning environment must include a strong element of peer review and expert critique.
- Efficiency and Focus are Paramount: Learners are acutely aware of time constraints. They want to "fast-track" knowledge and avoid "noodling" or getting distracted. This points to a need for highly curated, efficient, and goal-oriented learning paths.

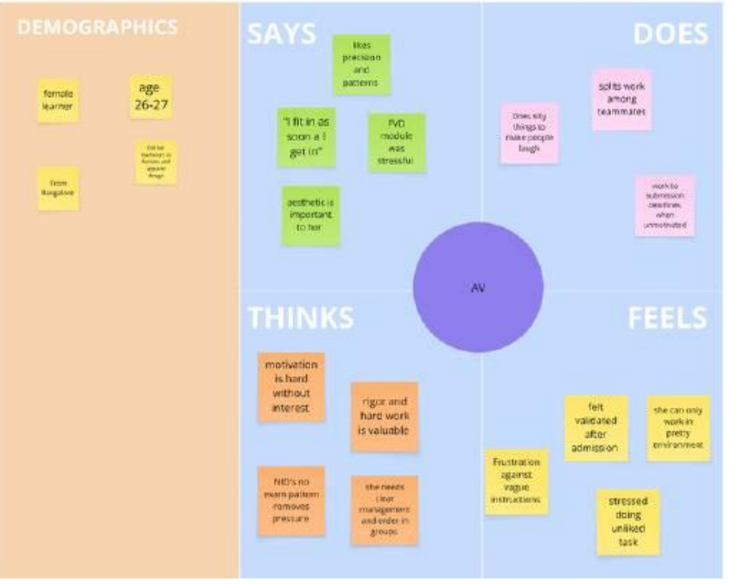
Subject 4



Subject 5

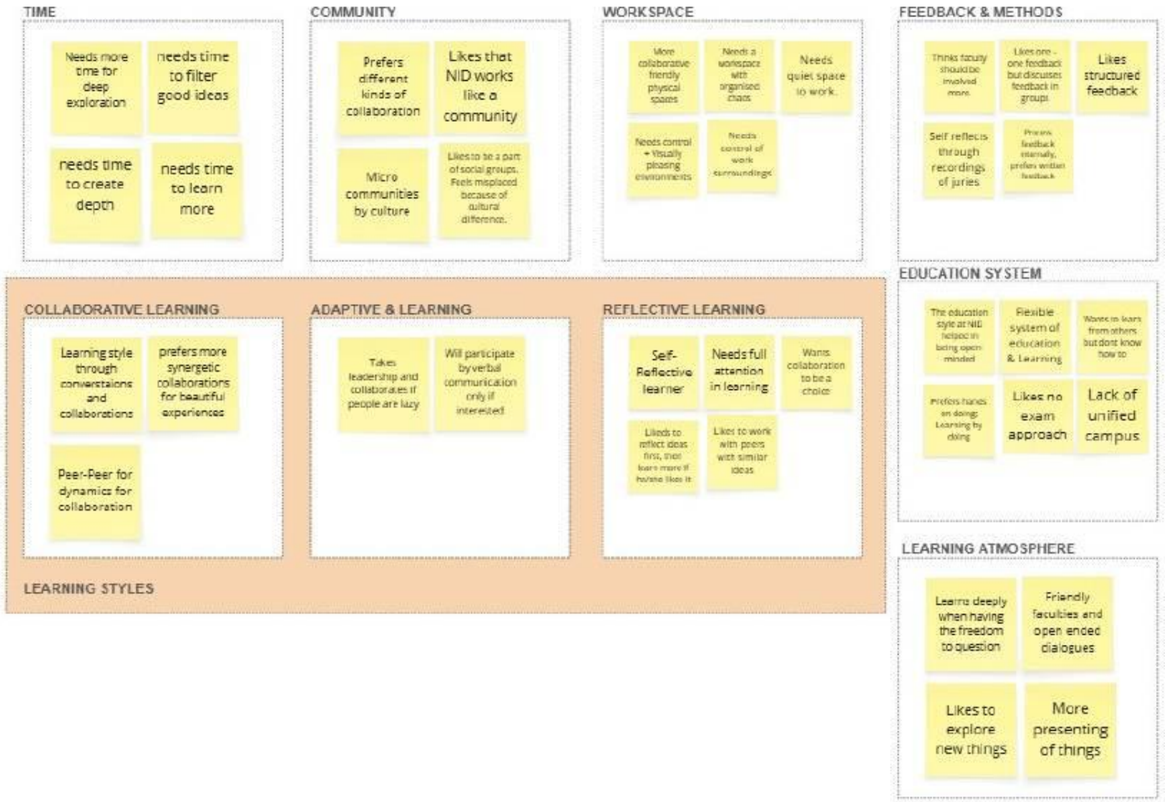


Subject 6



Affinity Mapping

1. The Universal Barrier: Overwhelm and Analysis Paralysis : Need for Structure: Learners across all demographics (Subjects 1, 2, 4) feel overwhelmed by the sheer volume of available information. They seek a "blueprint" or a simple, "fast-track" method to convert unstructured thinking into defined skills.
2. Actionable Starting Points: Users need clear, curated paths to overcome procrastination and the "Creative Block" (Subjects 2, 5, 6), indicating a strong demand for bite-sized content and project-based learning with explicit goals.
3. The Critical Need for Community and Validation
4. Structured Feedback Loop: Learners highly value structured feedback and expert critique (Subject 6, Affinity Map). They seek peer review and a system that offers validation to combat self-doubt (Subject 5).
5. Social Support: There is a persistent demand for a support system (e.g., a "3-tiered support group," "buddies" - Subject 1) and opportunities to connect with others (Subject 4) to share ideas and foster a collaborative learning atmosphere.
6. Focus on Practicality and Output
7. Learning by Doing: The priority is mastering skills through practical application and "learning by doing" (Subject 4, 3). Knowledge must be directly applicable to their professional goals.
8. Project Completion: Users struggle with finishing side projects (Subject 6). The system must be designed to help learners maintain focus and complete projects efficiently, translating inspiration into concrete output.
9. Demand for Control and Flexibility
10. Time Efficiency: Learners are highly conscious of time constraints (Affinity Map) and need a flexible system that allows them to mix and match modules, control their learning surroundings, and filter information efficiently.
11. Beyond Exams: There is a strong dislike for traditional, exam-driven learning and a desire for a flexible education system that focuses on self-reflection and allows the freedom to question and explore new things.



Part I: Discovery and Definition

The Causal Map

The Problem/Barrier

- Overwhelm & Paralysis: Learners are paralyzed by the volume of information and lack of a clear starting point.
- Time Constraint: Time for in-depth learning is the main inhibitor; efficiency and filtering are critical.
- Execution Gap: Users struggle with procrastination and converting ideas into finished projects.

Core Needs (The "What")

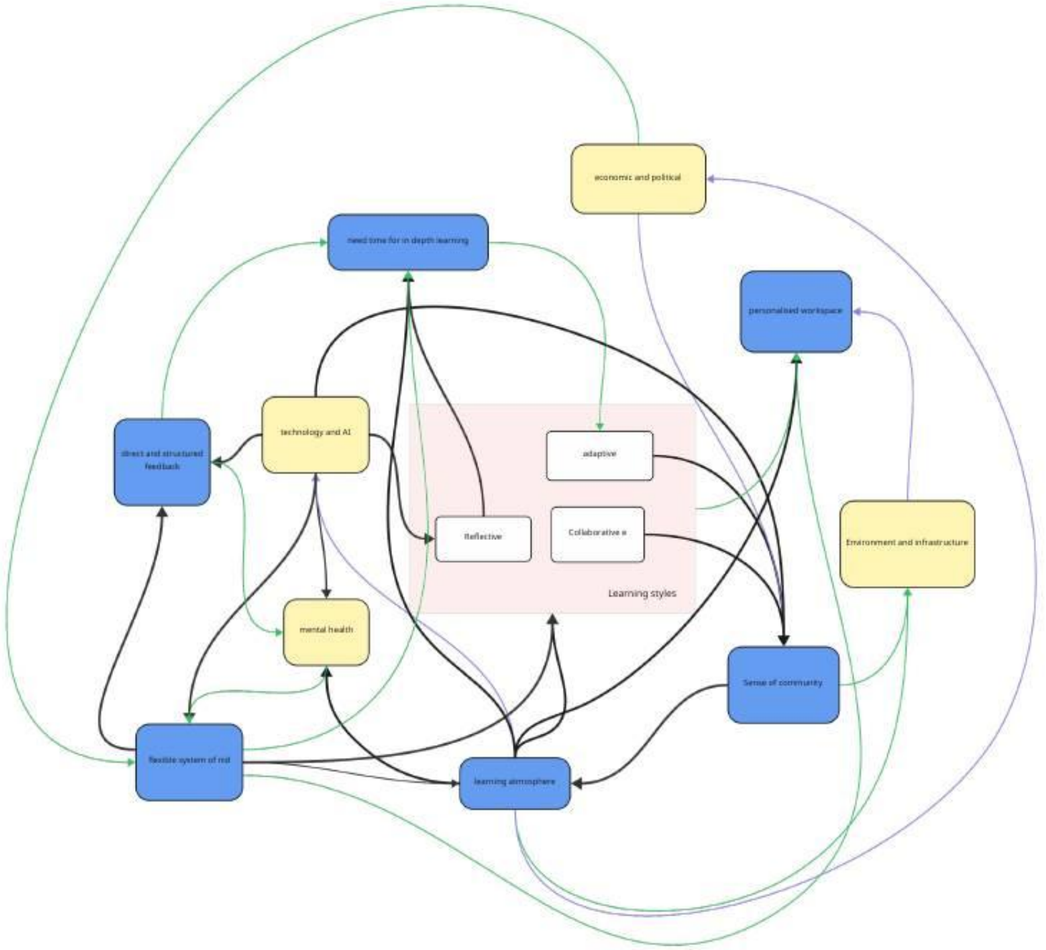
- Structured Path: Demand for a clear "blueprint" and simple, bite-sized entry points.
- Validation & Feedback: Crucial need for structured, expert feedback and peer review to combat self-doubt.
- Practical Application: Learning must be project-based and focus on mastering skills through "doing."
- Flexible System: The learning environment needs to be highly flexible (modules, control over workspace) and avoid rigid, exam-driven structures.

System Drivers (The "How")

- Adaptive Learning: Solutions must be adaptive to personalize the path and drive better feedback.
- Community Support: A strong social element (buddies, support groups, collaborations) is essential for motivation and validation.
- Reflective Practice: The system must facilitate reflective learning as it directly links to mental health and deeper understanding.

Key System Tension

- Personalization vs. Infrastructure: The system must be built with a flexible infrastructure (modules, workspace control) because rigidity is perceived as a barrier to personalized, comfortable learning.



Black Arrows: Strong Links
Green Arrows: Weak Links
Purple Arrows: Wrong Links

4.0 Design Brief & Problem Framing

1. Core Insights

Expression is fluid and situational - learners oscillate between reflection and articulation depending on context, confidence, and emotional safety.

Visibility ≠ Participation - quieter students often contribute meaningfully through making, writing, or observation, yet are overlooked in group settings that reward vocal presence.

Collaboration styles are asymmetric - some students take initiative to organize and structure, while others ideate freely; both roles are essential but unequally recognized.

Feedback timing matters - delayed or written feedback enables deeper reflection for some, while others thrive on instant, verbal critique.

Faculty and peers often misread quietness - silence is mistaken for disengagement rather than cognitive processing or careful observation.

Future learning needs adaptive collaboration - students seek environments and tools that dynamically adjust pace, interaction, and sensory input based on their expression style

2 . Research Question Iterations

How might we make collaborative learning better for individuals at NID in 2030?

▼

How do we make learning more collaborative for introverted individuals at NID in 2030?

▼

How do we make learning more collaborative for minimally expressive individuals at NID in 2030?

▶

How do we make learning more collaborative for individuals with different levels of expression at NID in 2030?

3 . Design Goals

Design Brief: To design and develop solutions promoting collaborative learning among different styles of learners within NID in 2030

Promote empathy and awareness across expression styles.

Encourage learners to recognize that everyone contributes differently — some think out loud, others through quiet making or observation. By building empathy for these differences, collaboration becomes more patient, inclusive, and human.

Design systems that enable balanced collaboration.

Create tools, platforms, or environments that naturally balance expressive and reflective energies within a group. The aim isn't to make quieter learners more vocal, but to give every archetype space to contribute in the way they do best — and be valued for it.

Make invisible participation visible.

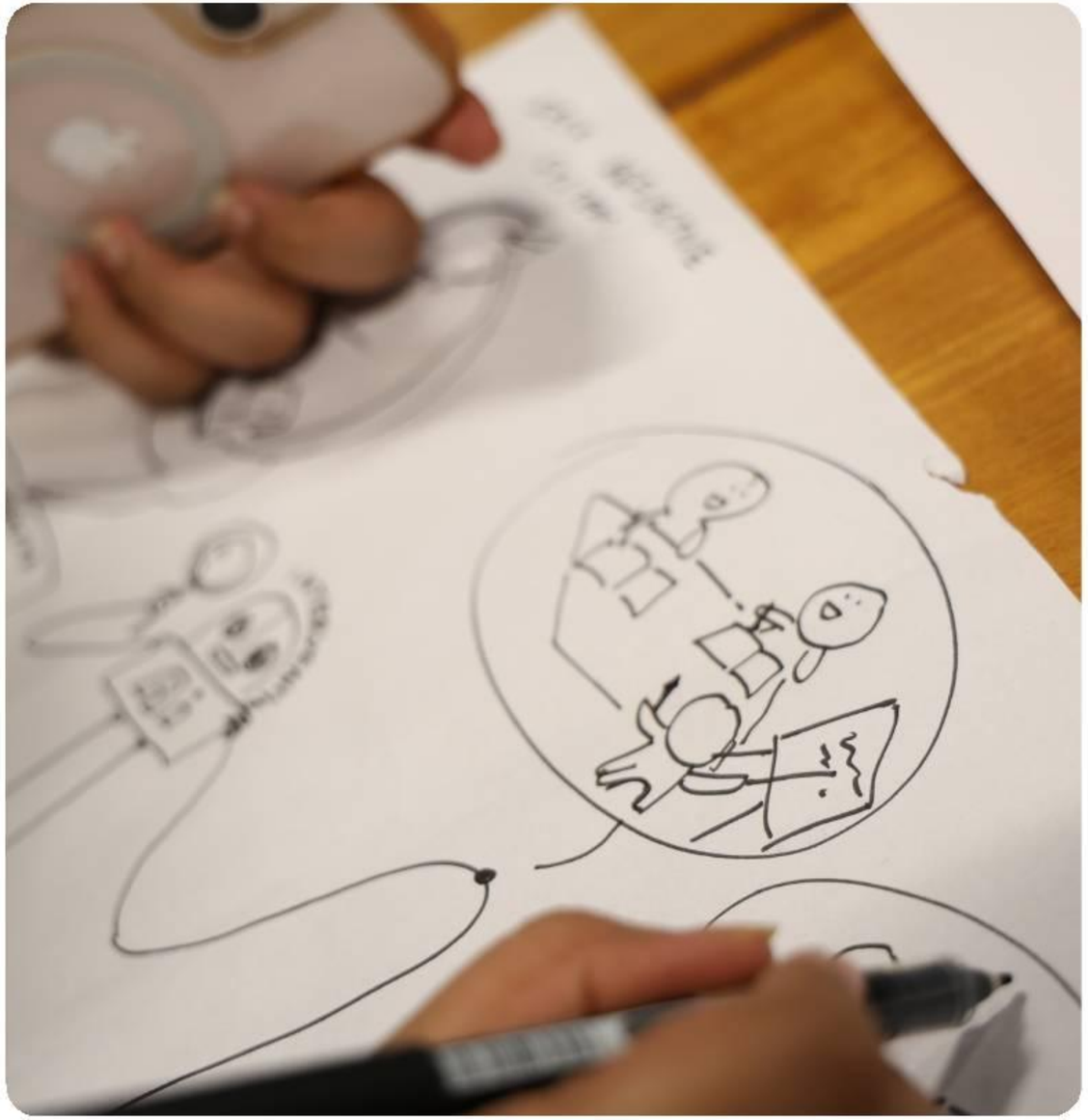
Redefine what "participation" means by acknowledging listening, synthesis, and behind-the-scenes effort as equally impactful forms of engagement. These quieter contributions need to be surfaced through subtle design cues and feedback loops.

Ensure every learner feels heard and effective.

Collaboration should lead to mutual growth, where expressive learners learn to pause and listen, and reflective learners feel empowered to share without pressure. In the end, the collaboration should feel fruitful for all, not dominated by one energy type.

Develop flexible evaluation metrics for diverse archetypes.

Each archetype measures success differently — a facilitator thrives on group harmony, while a thinker values conceptual clarity. We realized that our testing couldn't fully quantify this diversity, but even intuitively, it was clear that one fixed metric can't define collaborative success.



Ideation & Development

5.0 Ideation & Concept Selection
6.0 Prototyping & Iteration
7.0 Final Design & Selected theme

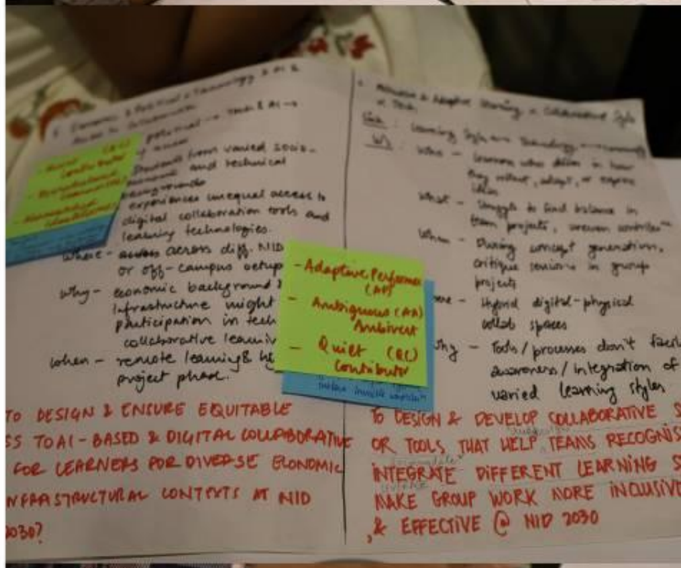
5.0 Ideation & Concept Selection

1. Ideation Methods

We used bodystorming, roleplaying, “Wizard of Oz” testing, and How Might We mapping to explore collaborative scenarios between different learner archetypes.



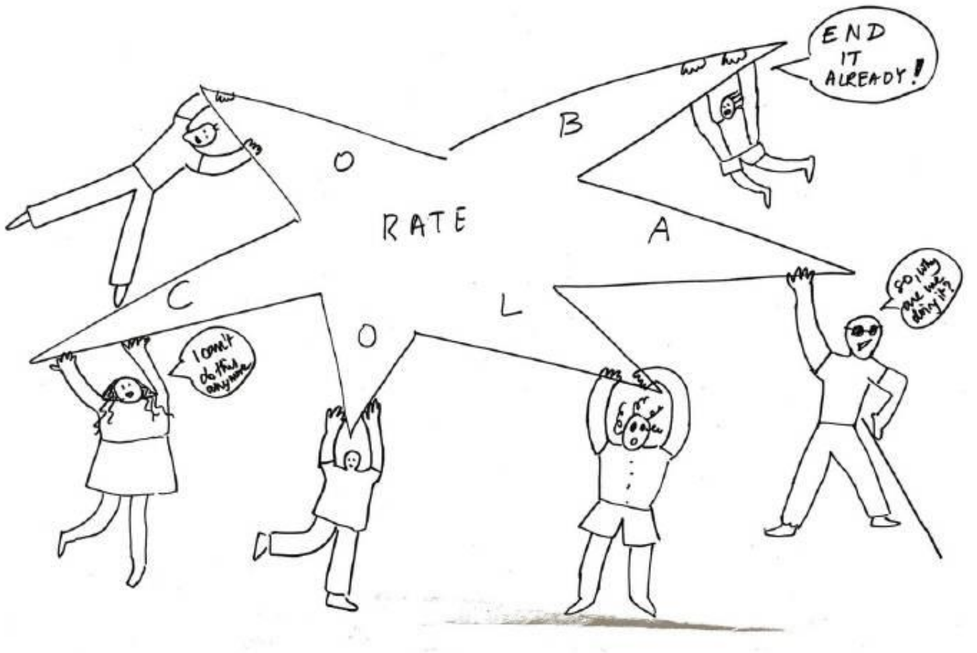


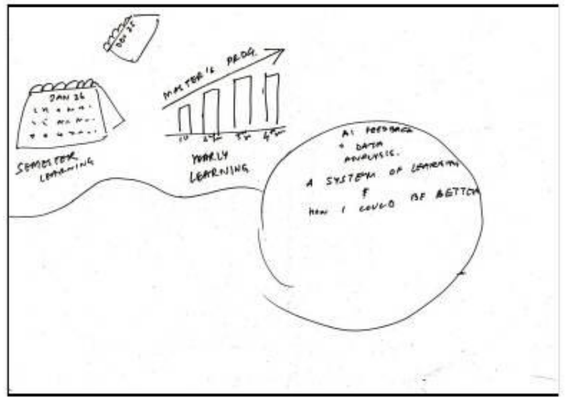


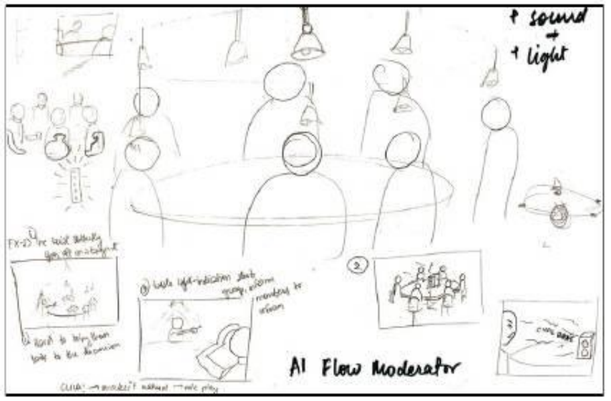


2. Initial Concepts

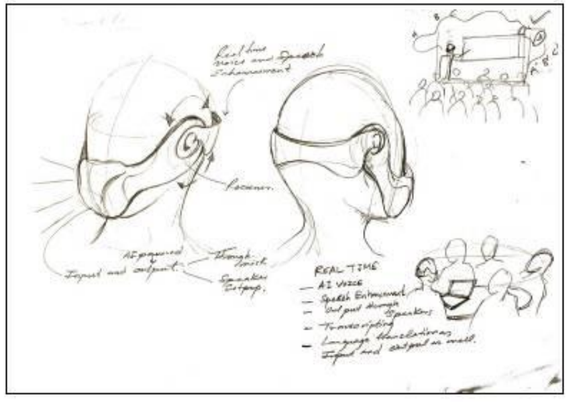
1. Data-driven algorithmic grouping to form groups
2. A collaboration platform for deep learning (NID Commons)
3. DIN-RAAT AI (AI substitute for mentors at night time)
4. A collaborative learning policy system (embed it in the teaching system)
5. Choose your mentor
6. Self-reflective system
7. Scenario based RPGs
8. Energy Reflector Wearables
9. Beehive Modular Partitions
10. AI Persona substitute
11. Digital Articulation System - extension or device
12. AI Energy Flow Mediator - Adaptive Lighting System/Voice assistant

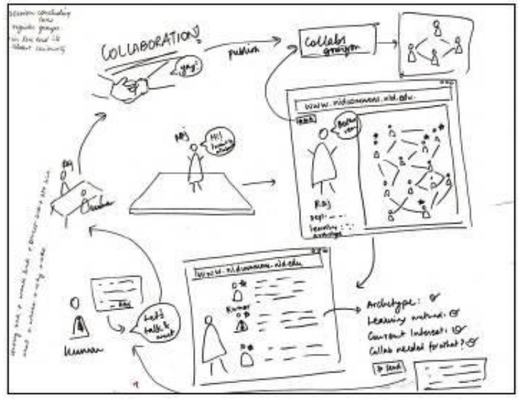


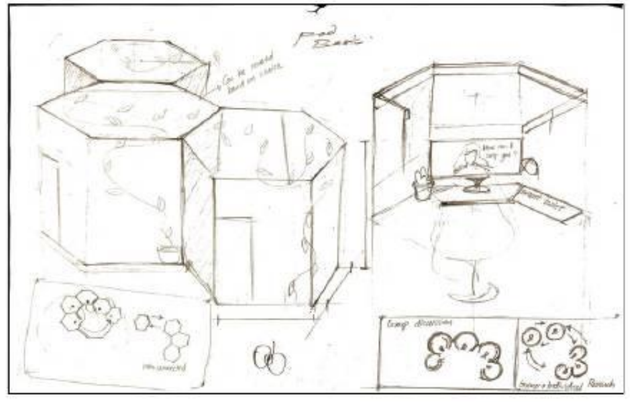

SEMESTER LEARNING
WEEKLY LEARNING
AI FEEDBACK + DATA ANALYSIS.
A SYSTEM OF LEARNING
&
HOW I COULD BE BETTER

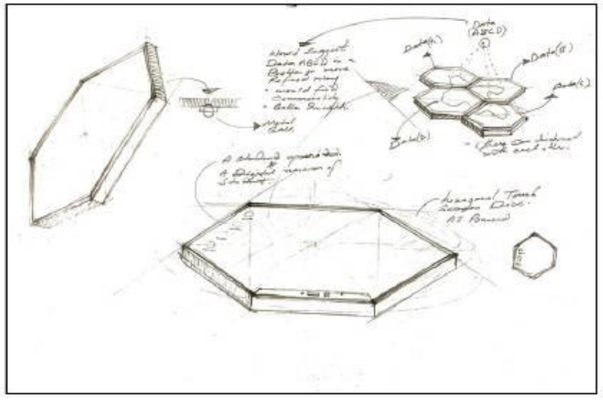

+ sound
+ light
AI Flow Moderator

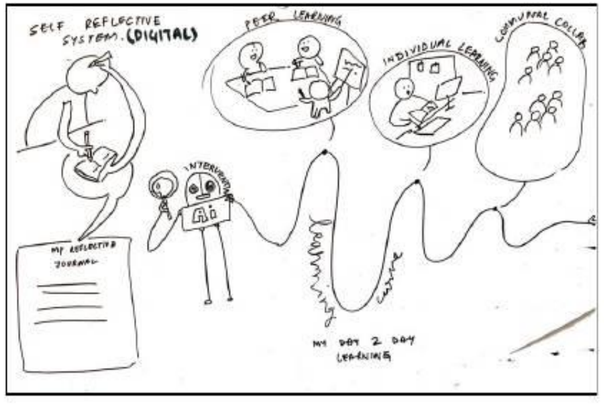


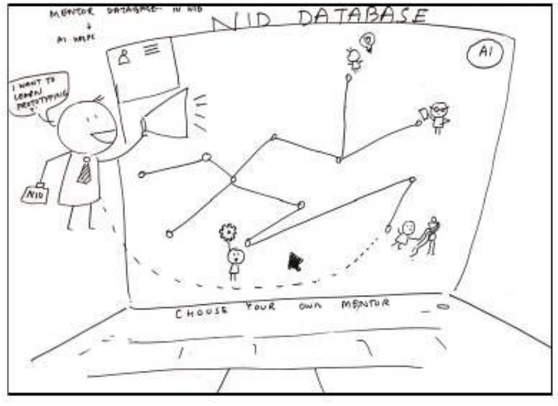

Real time voice and Speech Enhancement
Facemesh
AI powered Input and output
Through text
Through voice
Speaker Output
REAL TIME
- AI VOICE
- Speech Enhancement
- Output through Speakers
- Transcripting
- Language Translating
- Input and output in real.


COLLABORATION
publish
Collab
Archetypes:
Learning method:
Content interest:
Collab needed format?
Human


Feed Back
Group discussions


Hexagonal Tomb
AI Bound


SELF REFLECTIVE SYSTEM. (DIGITAL)
PEER LEARNING
INDIVIDUAL LEARNING
COMMUNITY COLLAB
INTERVIEWER
MY REFLECTIVE JOURNAL
learning curve
MY DAY 2 DAY LEARNING


MENTOR DATABASE IN NID
AI HELPS
NID DATABASE
AI
I WANT TO LEARN PROTOTYPING
CHOOSE YOUR OWN MENTOR


GOAL: Finish the day with everyone thriving :)
or studio collapses from burnout !
The 'inclusive' STUDIO
ENERGY TOKEN
COLLAB TOKEN
STRESS TOKEN
START
BRAINSTORM
IDEATION
PROTOTYPE
CRITIQUE
HAPPIE ZONES
CARD DECKS
TASK CARDS
STUDIO EVENT CARDS
The 7 Archetypes

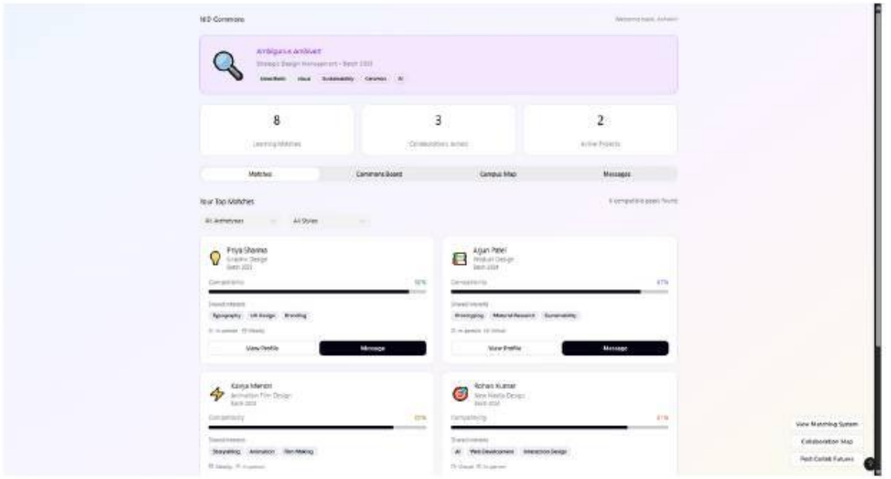
3. Concept Selection Logic:

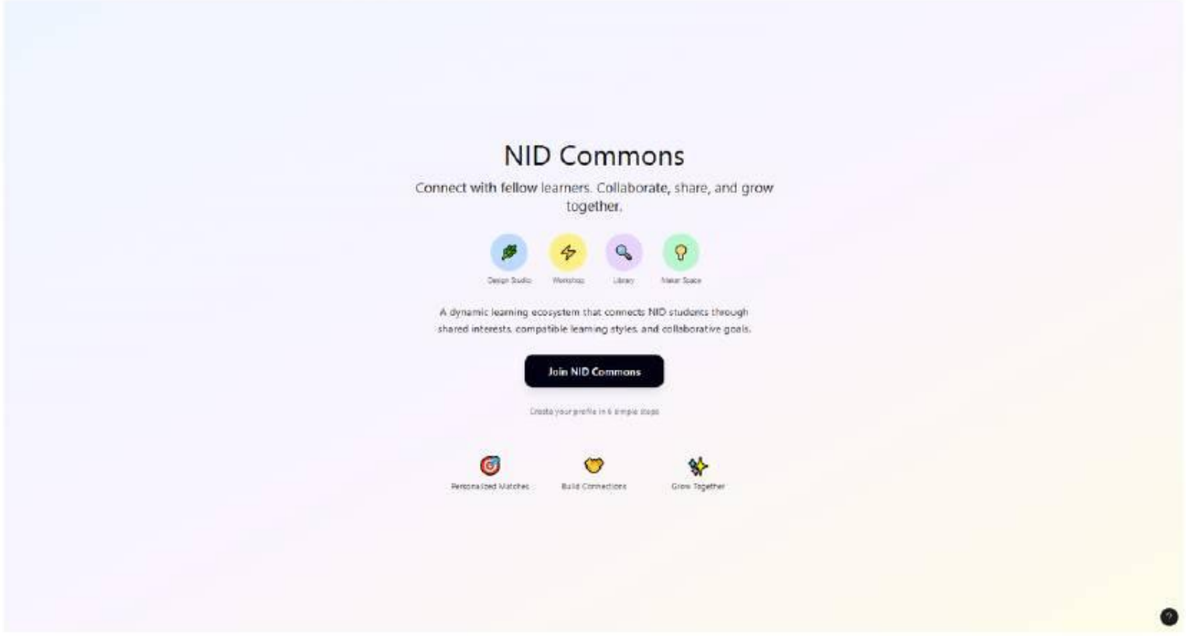
Concept Idea	Plausibility (Feasibility in 2030 context)	Impact (Potential to improve collaboration & inclusivity)	Experimentation (Novelty & exploratory potential)	Selected
Data-driven algorithmic grouping	★★★★☆ – Technically plausible; aligns with AI-personalization trends.	★★★☆☆ – Risk of over-automation and loss of human agency.	★★★☆☆ – Familiar predictive model; less creative experimentation.	❌
NID Commons – Collaboration Platform for Deep Learning	★★★★★ – Scalable, aligns with hybrid collaborative learning ecosystems.	★★★★★ – Encourages cross-disciplinary, expressive-reflective collaboration.	★★★★☆ – Allows flexible UX experimentation and identity design.	✅
DIN-RAAT AI Mentor	★★★★☆ – Technically feasible as conversational AI; fits 24×7 studio rhythm.	★★★☆☆ – Helpful but limited to guidance, not emotional collaboration.	★★★★☆ – Fun speculative narrative, but less collaborative in core.	❌
Collaborative Learning Policy System	★★★☆☆ – Institution-level reform, not a designed product.	★★★★☆ – High systemic impact but low immediacy.	★★☆☆☆ – Abstract; low creative prototyping potential.	❌
Choose Your Mentor	★★★☆☆ – Feasible socially, not infrastructurally.	★★★★☆ – Improves self-direction and comfort.	★★★☆☆ – Moderate room for creative framing.	❌
Self-Reflective System	★★★★☆ – Simple and feasible, integrates well into existing learning models	★★★☆☆ – High individual but low collective impact.	★★★★☆ – Rich for speculative UX and psychological cues.	❌
Scenario-based RPGs	★★★★☆ – Easy to simulate digitally or physically.	★★★★☆ – Excellent for empathy-building and	★★★★★ – Playful, immersive, and speculative.	✅
Energy Reflector Wearables	★★☆☆☆ – Low practical feasibility and maintenance	★★★☆☆ – Conceptually interesting but limited to	★★★★☆ – Experimental but peripheral to collaboration.	❌
Beehive Modular Partitions	★★★★☆ – Spatially plausible; low-tech and sensory driven.	★★★☆☆ – Improves environment indirectly; low behavioral change.	★★★☆☆ – Moderate spatial experimentation.	❌
AI Energy Flow Mediator (Adaptive Lighting + Voice Assistant)	★★★★★ – Plausible with emerging adaptive systems.	★★★★★ – Balances team dynamics and regulates group energy.	★★★★★ – Integrates spatial, AI, and behavioral experimentation.	✅

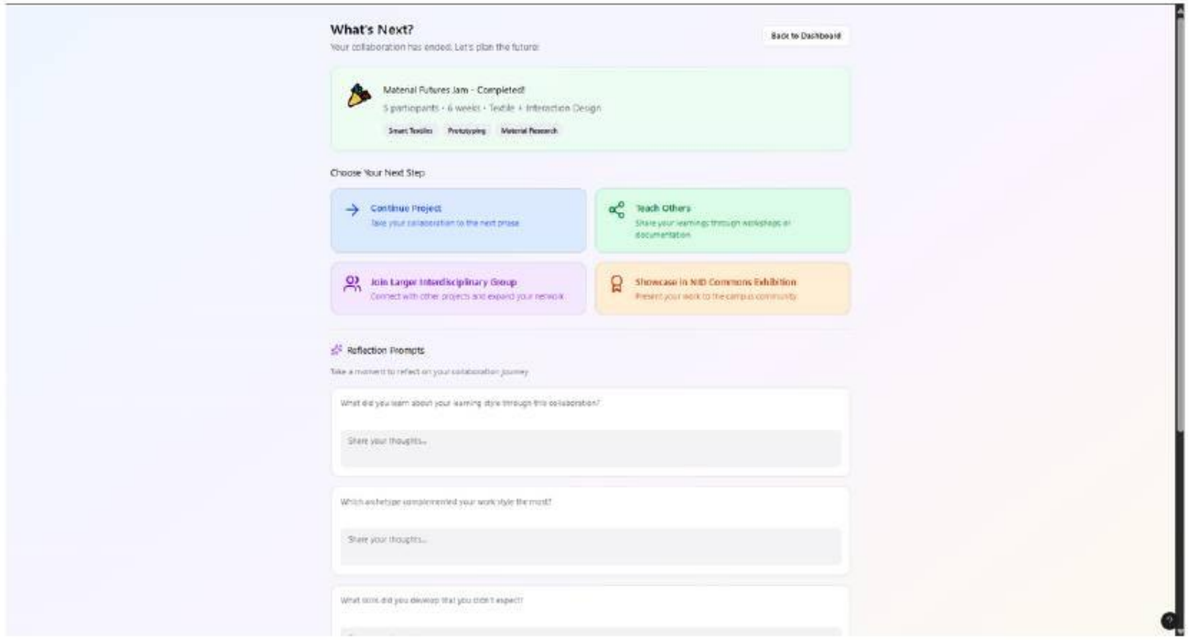
6.0 Prototyping & Iteration

Prototype 1.0 (basic functions)





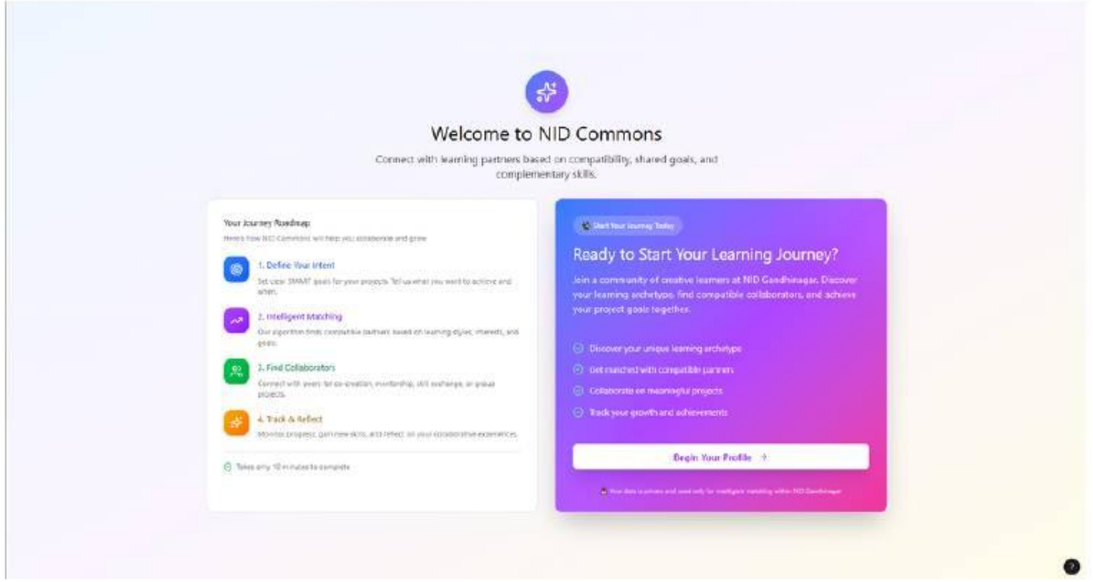




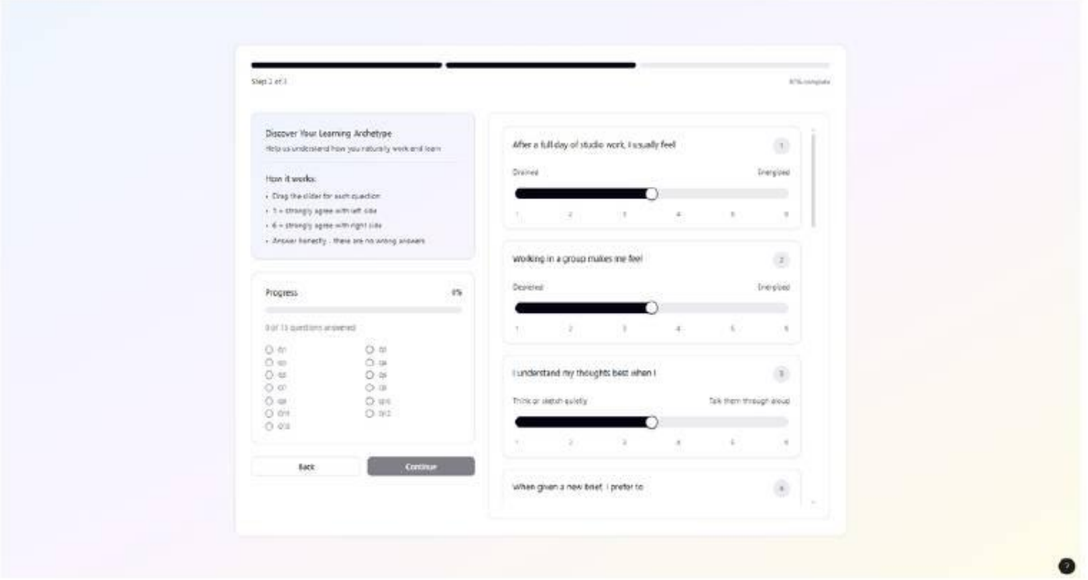
This platform just matched the users on the basis of all our research to make a compatibility matrix. Major setback was this would not be helping collaborations when asynchronised learners came together.

6.0 Prototyping & Iteration

Prototype 1.1



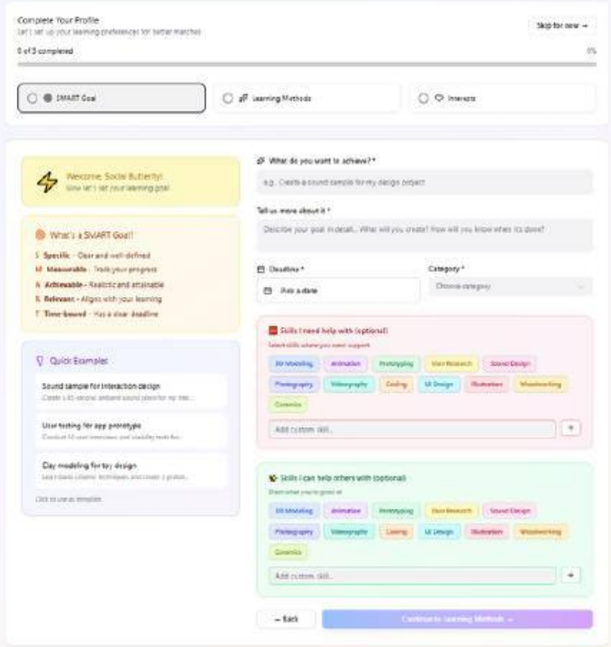


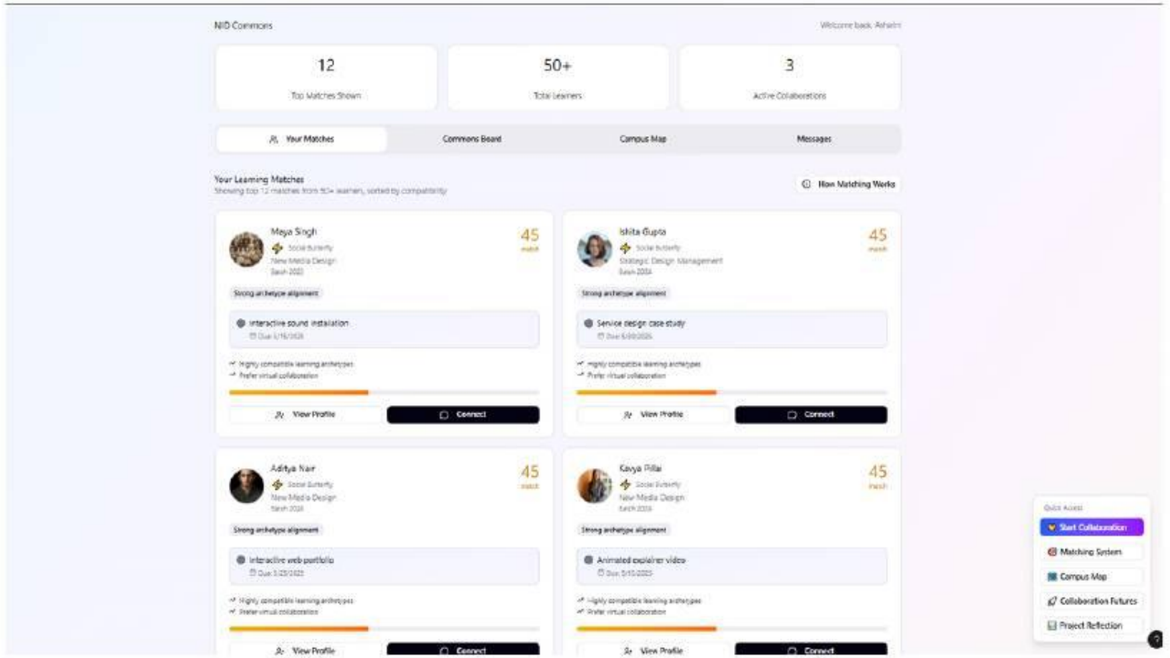


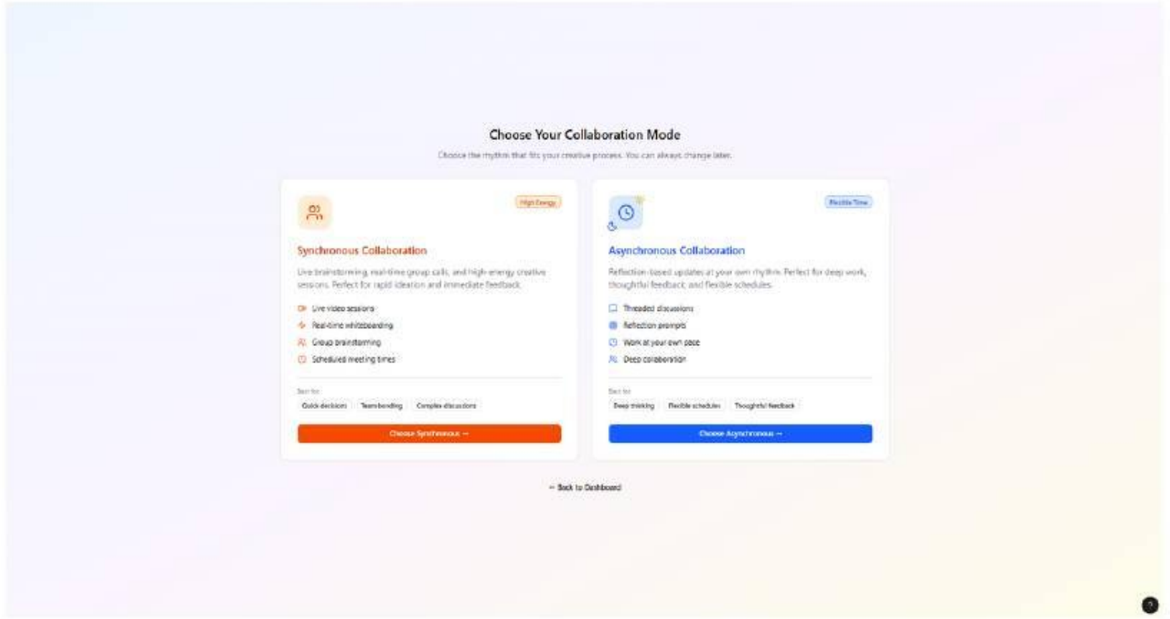
This version of the prototype used SMART Goals Framework to solidify the matrix and provide better compatibility and also help asynchronised learners.

6.0 Prototyping & Iteration

Prototype 1.1

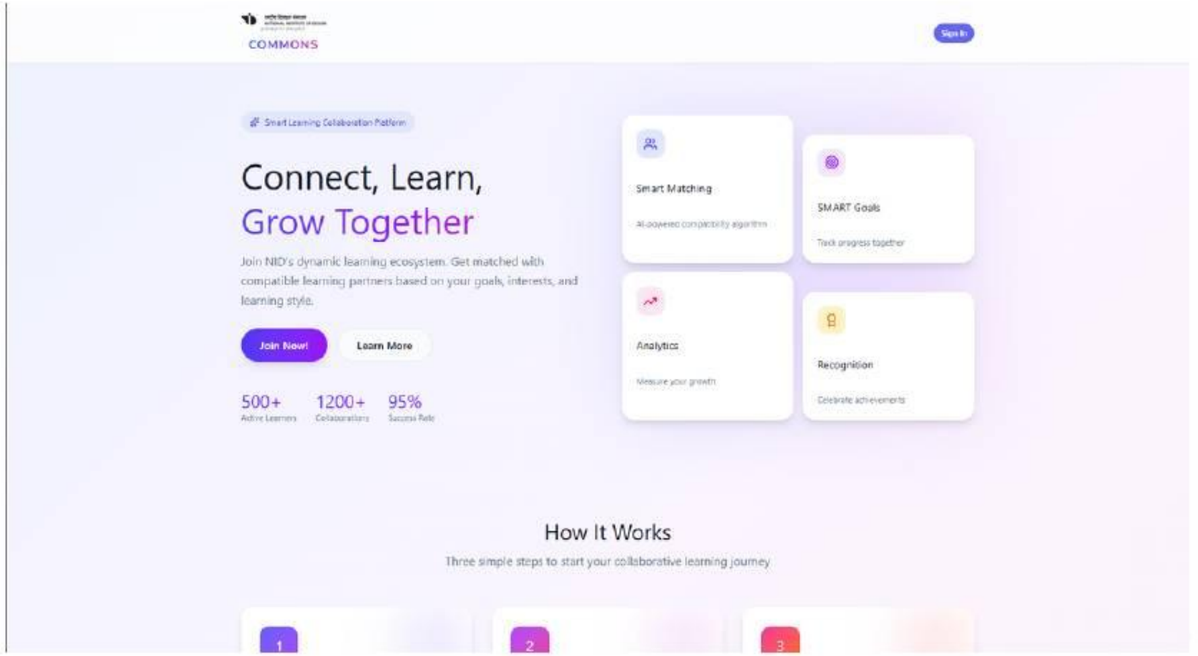




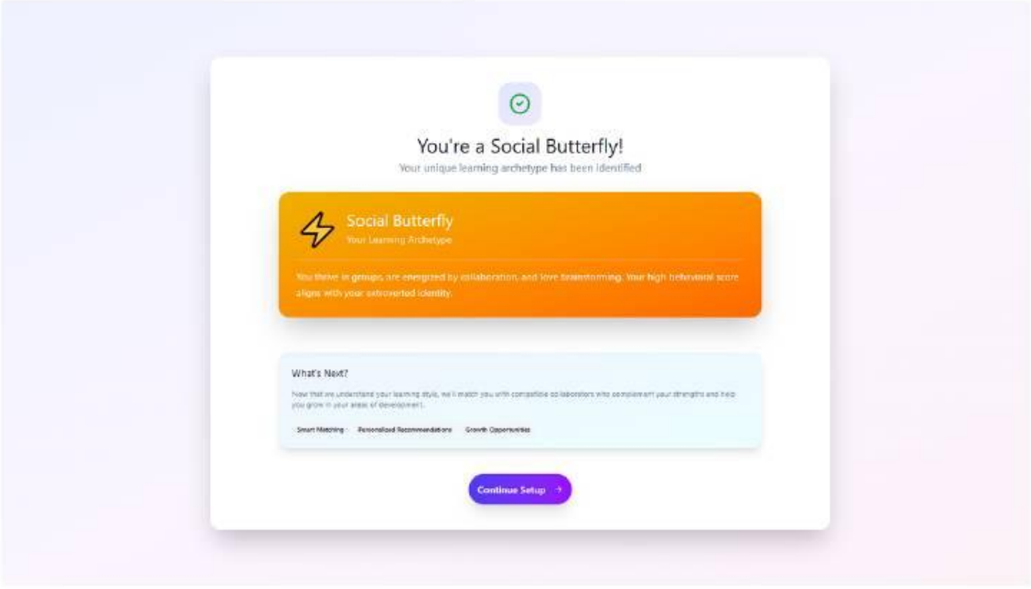


6.0 Prototyping & Iteration

Prototype 1.2



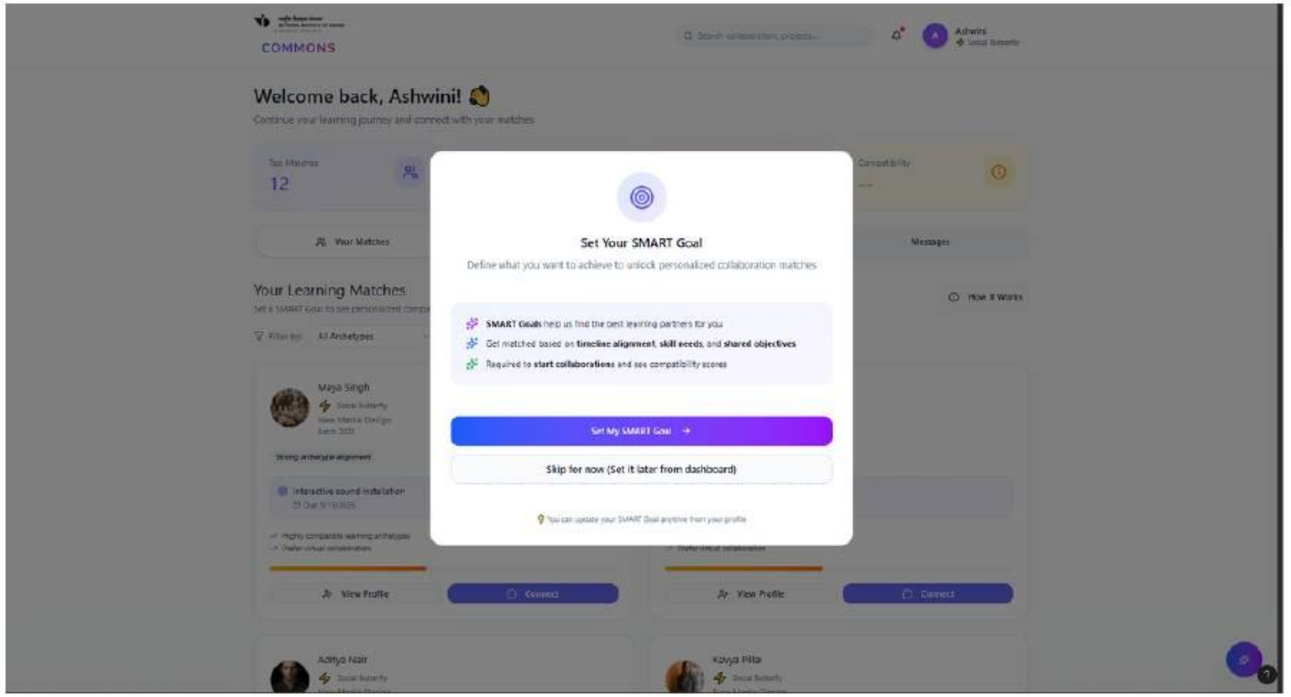


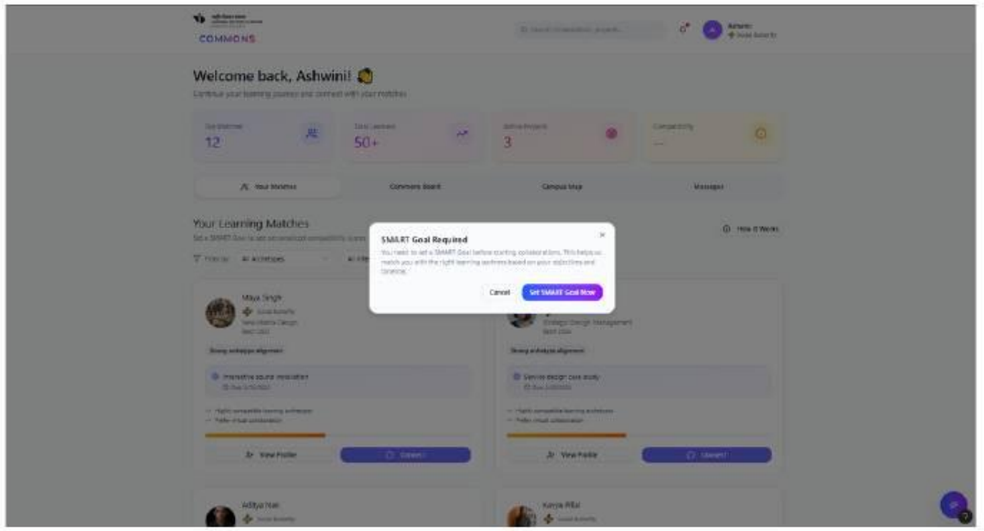


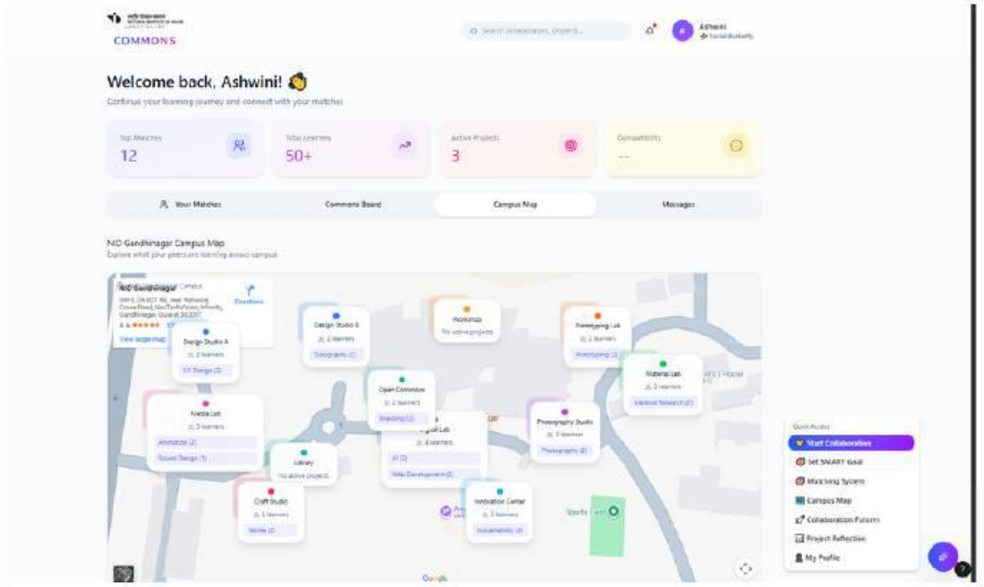
The overall UI and user flow was enhanced and made sure it is easy and inclusive to use.

6.0 Prototyping & Iteration

Prototype 1.2



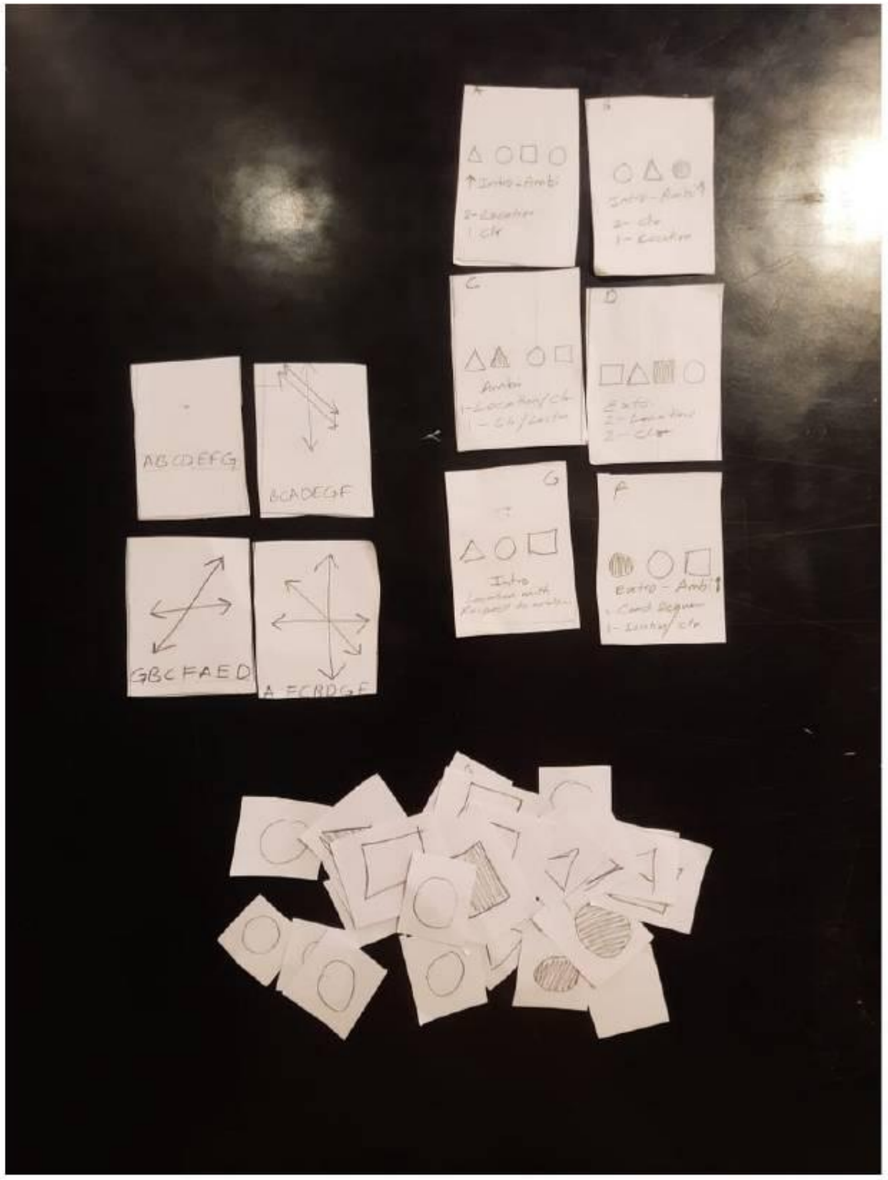




Unless a learner came to the platform came with a purpose, collaboration was halted. Additionally, the campus map was incorporated to show all collaboration spots.
Final URL: https://amount-hatch-73522679.figma.site

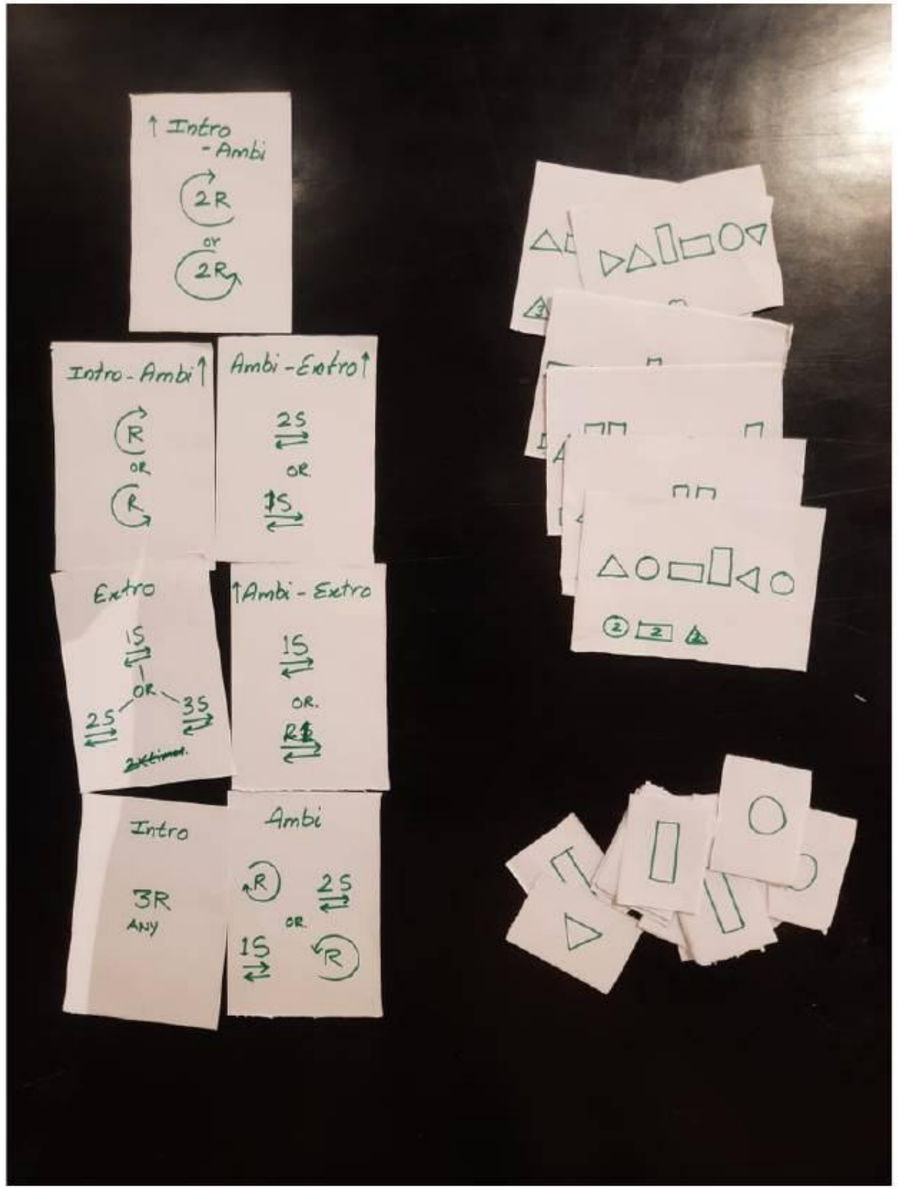
6.0 Prototyping & Iteration

Prototype 2.0



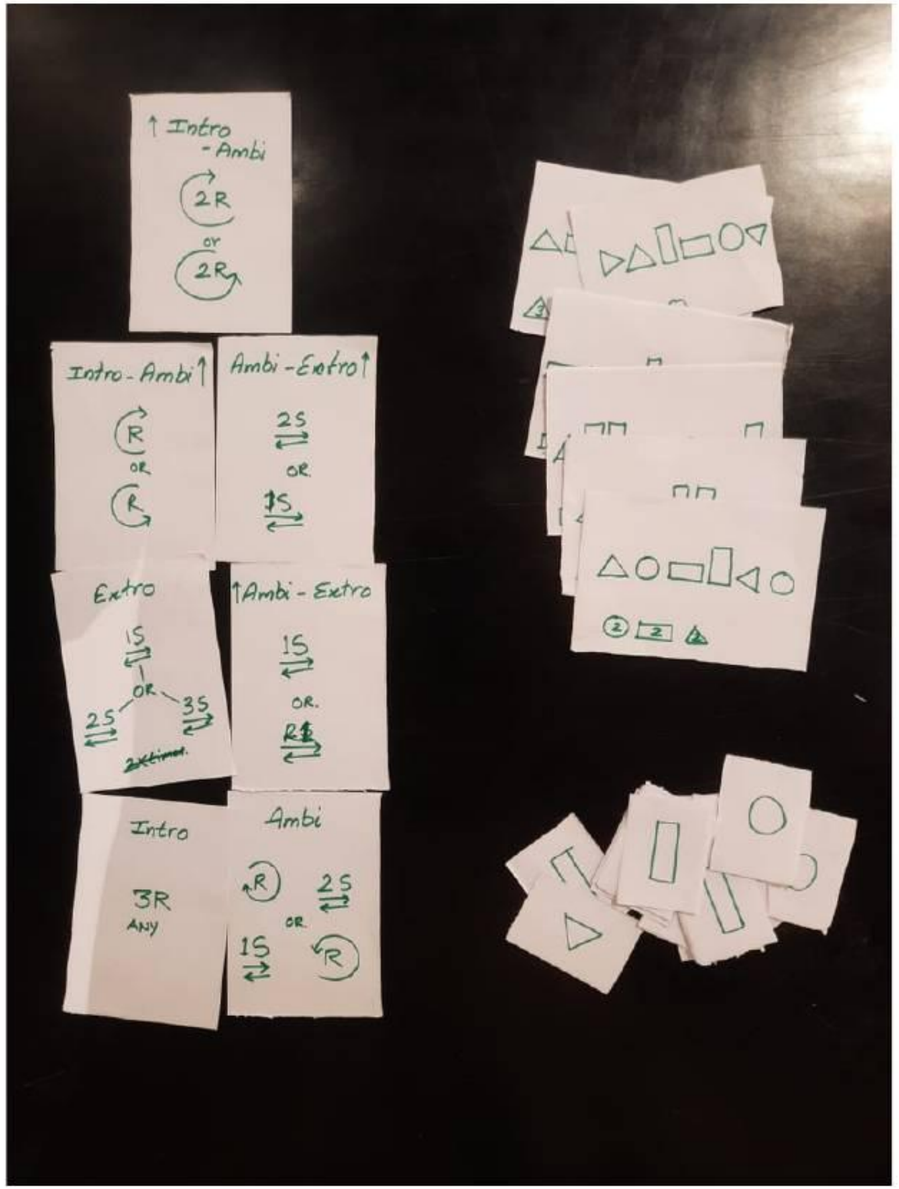
6.0 Prototyping & Iteration

Prototype 2.1



6.0 Prototyping & Iteration

Prototype 2.1



6.0 Prototyping & Iteration

Prototype 2.2

The game structure is designed to improve group collaboration by forcing players to practice three critical skills:

1. **Limited & Effective Communication:** Since you can't reveal the goal pattern, you must use precise, non-redundant language to describe actions and exchanges only. This hones the skill of conveying critical intent with minimal input.
2. **Shared Mental Model:** Players must build a shared understanding of the hidden goal through subtle cues and the effects of each other's actions, which fosters collective strategizing.
3. **Mutual Dependency:** Success hinges on relying on a teammate's specific action (Move vs. Rotate) at a specific time. This builds trust in each other's specialized roles and promotes effective decentralized control.



Game Objective
Players must recreate the specific sequence or pattern shown on the Challenge Card by strategically moving Shape Tokens across the board.
To do so, each player will utilize the unique abilities granted by their assigned Archetype Card.

Setup
Shuffle the 7 Archetype Cards and distribute one to each player. Each card represents a distinct archetype with a unique power or ability.

Place the Challenge Card Deck face down at the center of the play area. Draw the top card to reveal the pattern that all players must replicate

stack the shape tokens

Gameplay
There are no turns — all players act simultaneously, racing to complete the challenge pattern.

Each player can use the ability on their Archetype Card to move or manipulate the tokens.

Arrows on the Archetype Card indicate the direction of movement or rotation:
Clockwise (↻) or Anti-clockwise (↺).

The terms Swap, 2x, or 3x describe movement or exchange actions:
Swap → exchange two cards or tokens.

2x / 3x → indicates the number of cards or spaces between swapped items.

Rules
Each Archetype Card has a unique ability — use it wisely.
No undoing moves once made; every action is final.

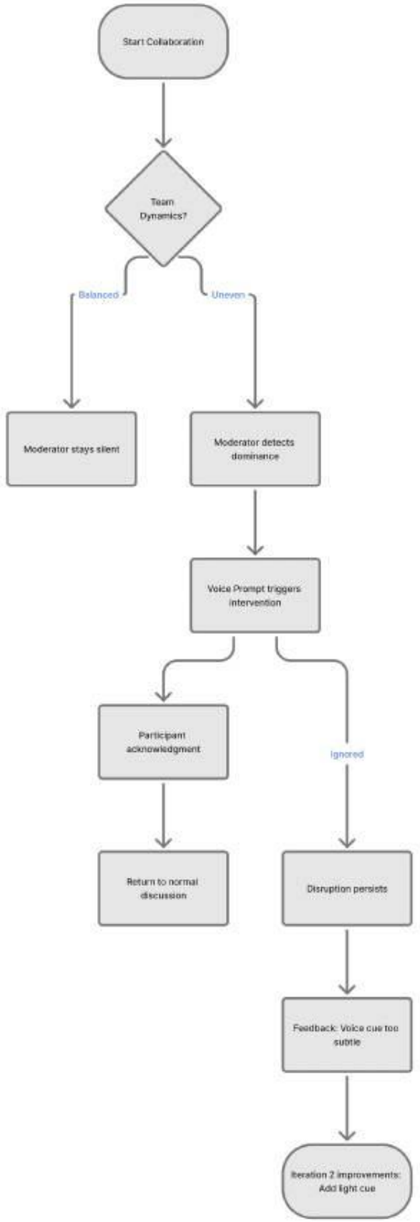
Collaboration and strategy are key — since all players move at once, timing and observation matter.

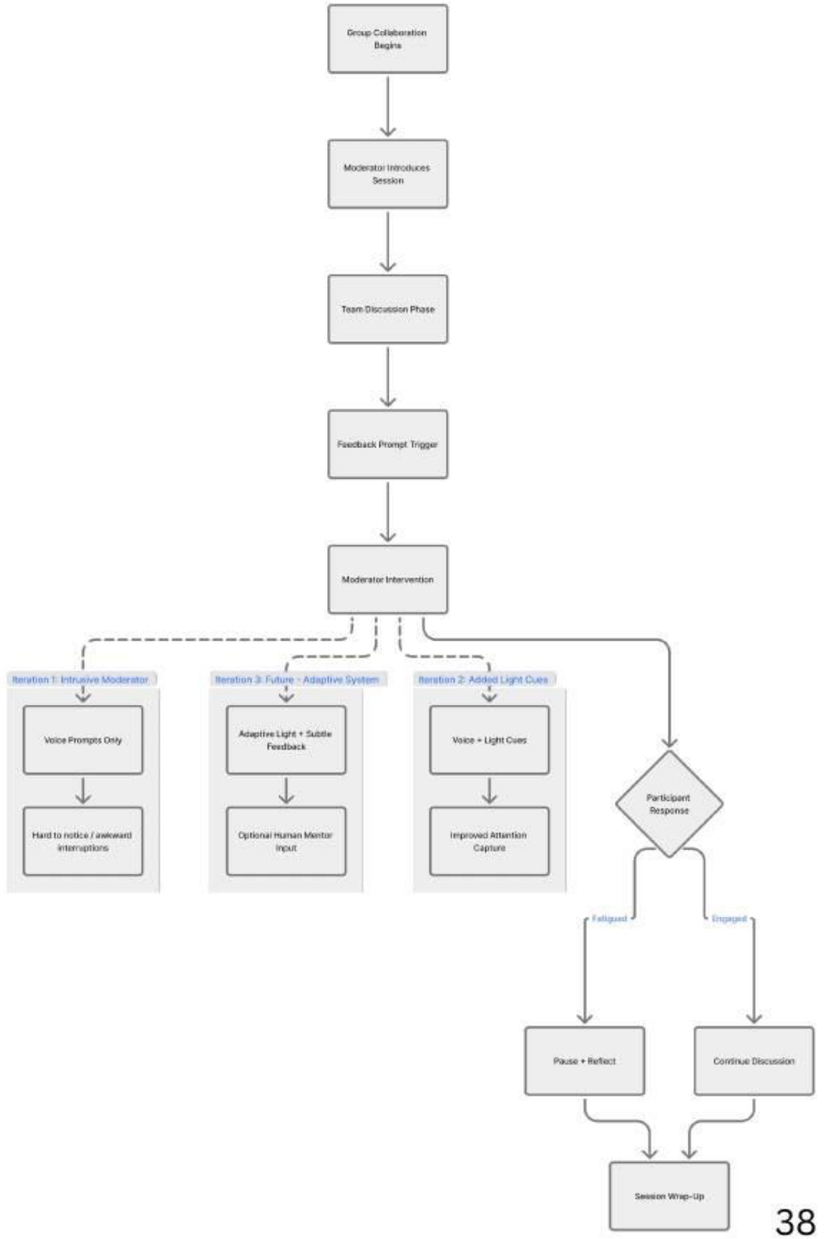
Design Insight Summary
This game combines pattern recognition, real-time strategy, and character-driven abilities to encourage both individual creativity and collective chaos Simultaneous play removes downtime and creates dynamic tension. The movement mechanics (rotation, swapping, sequencing) engage both spatial reasoning and tactical decision-making



6.0 Prototyping & Iteration

Iterations for the Scenario Flow





6.0 Prototyping & Iteration

Prototype 3.0



6.0 Prototyping & Iteration

Prototype 3.1



6.0 Prototyping & Iteration

Prototype 3.2



Part III

Testing & Reflection

8.0 Final Validation
9.0 Conclusion & Learning



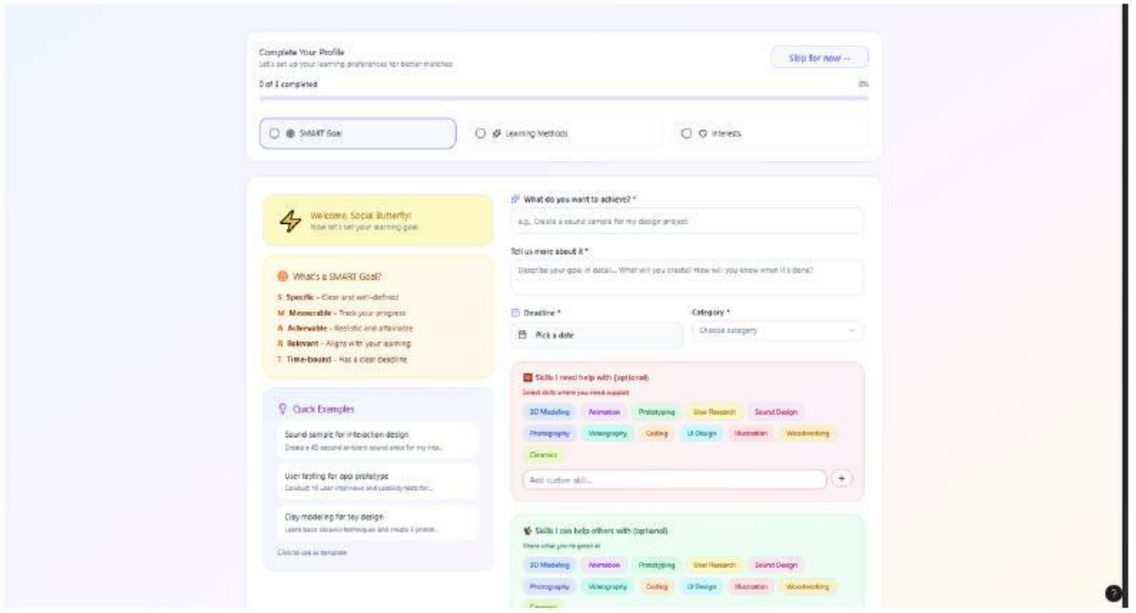
7.0 Final Validation

Final Test Methodology

Prototype 1

Three iterative prototypes were tested to evaluate inclusivity, engagement, and collaboration across diverse learner archetypes. Prototype 1.0 used an algorithmic compatibility matrix to form balanced teams; while it visualized learner diversity, it failed to sustain engagement among reflective participants. Prototype 1.1 integrated the SMART Goals framework, improving clarity, accountability, and participation balance, yet collaboration still depended on user initiative. Prototype 1.2- the NID Commons platform - combined digital collaboration, adaptive campus mapping, and reflective interaction tools. Overall, the evolution from algorithmic grouping to adaptive ecosystems demonstrated how structured flexibility fosters deeper, more balanced collaboration at NID 2030.



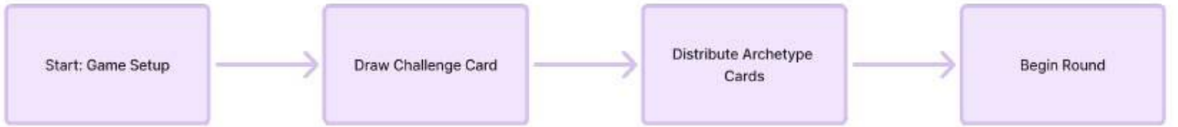


7.0 Final Validation

Final Test Methodology

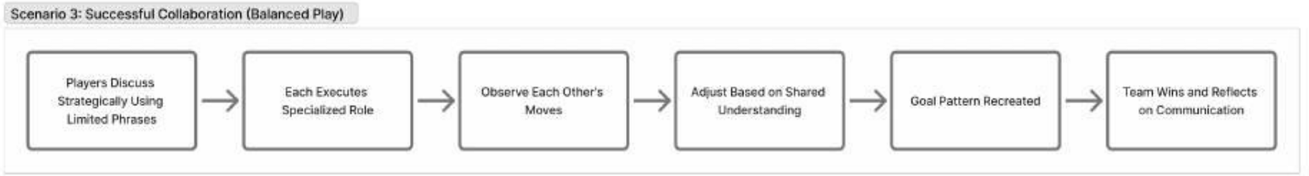
Prototype 2

Scenario-based testing with observational analysis. Teams engage in a structured game round, and their interactions are mapped against three predefined behavioral outcomes:
Communication Breakdown (Team Loses)
Overpowered Player (Team Wins, but Collaboration Fails)
Successful Collaboration (Team Wins and Learns)
Why Used: To directly observe and validate how game mechanics influence specific collaborative skills (limited communication, shared mental model, mutual dependency) and to understand the causes of success or failure in group dynamics. It reveals both effective and detrimental interaction patterns.
Key Insights Gained:
Uncoordinated action without communication leads to guaranteed failure.
Single-player dominance undermines learning, even if the goal is achieved.
Strategic, limited communication, specialized roles, and adaptive understanding are essential for successful, reflective collaboration.

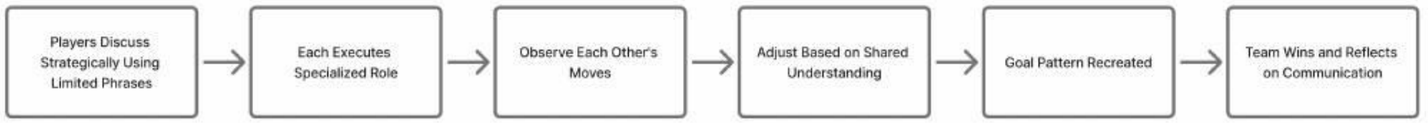








Scenario 3: Successful Collaboration (Balanced Play)



Scenario 2: Overpowered Player (Unbalanced Play)



Scenario 1: Breakdown of Communication (Team Loses)



7.0 Final Validation

Final Test Methodology

Prototype 3

The Wizard of Oz AI prototype simulated an intelligent mentor to explore how conversational assistance could support learners during collaborative sessions. Using a controlled mock interface, human facilitators secretly operated the “AI” to mimic real-time guidance, emotional check-ins, and reflective prompts. Ten students across varied archetypes interacted with the system while working on an open brief, believing it to be autonomous. The session measured response relevance, emotional tone, and user comfort in sharing personal reflections. Reflective learners appreciated the non-judgmental, text-based dialogue for processing ideas privately, while expressive users valued quick motivational feedback. However, overreliance on automated empathy and reduced peer interaction emerged as risks. Observations suggested that AI can supplement—but not replace—human mentorship, functioning best as a reflective companion within a broader collaborative system. The test confirmed the potential of emotionally aware AI to enhance balance and psychological safety in hybrid learning.



8.0 Conclusion & Learning

Project Reflection

The project journey from empathy research to speculative prototyping illuminated how collaboration is as much an emotional process as it is an academic one. By mapping learner archetypes and decoding the invisible spectrum between reflection and expression, our team gained a nuanced understanding of inclusivity in design education.

Developing NID Commons revealed the complexity of designing for human diversity — the challenge was not to “fix” introversion or amplify extroversion, but to design conditions where every learner feels seen, valued, and effective. Through iterative prototyping, we learned the importance of feedback timing, visible yet non-intrusive participation cues, and adaptable learning rhythms.

Testing underscored that technology can facilitate connection but cannot replace human empathy. True collaboration emerged when systems allowed for autonomy, reflection, and organic engagement. This insight reframed our definition of success — from building a perfect platform to initiating a cultural shift toward patience, inclusivity, and balanced expression.

Ultimately, Collaborative by Design reaffirmed that the future of learning lies not in uniformity, but in designing environments that celebrate difference — where silence, speech, and making all coexist as equal forms of participation.

Part III: Testing & Reflection (The "Impact")

Future Recommendations

- Scalability Beyond NID: The systems and prototypes developed — particularly NID Commons — can be adapted to other educational institutions by creating open-source modules. Future iterations should explore interoperability with existing LMS (Learning Management Systems) and hybrid collaboration tools to expand accessibility and cross-campus co-learning.
- AI Ethics and Transparency: As AI-mediated collaboration becomes integral to learning, future research should focus on establishing ethical frameworks for algorithmic grouping, authorship recognition, and emotional data use. Transparency in how AI assesses compatibility and feedback is vital to preserve trust and agency among learners.
- Longitudinal User Testing: A semester-long pilot involving diverse learner archetypes would offer deeper insight into behavioral shifts, engagement patterns, and cognitive load. Tracking participation, reflection frequency, and project completion rates over time can quantify inclusivity and impact more accurately.
- Physical-Digital Synergy: The prototypes should evolve to integrate adaptive spatial environments—such as smart studios that respond to energy rhythms and collaboration intensity. Merging environmental cues (light, sound, spatial density) with digital data can create holistic, sensory-balanced learning experiences.
- Faculty Development and Policy Integration: To embed inclusivity systemically, future work must focus on faculty sensitization workshops and policy interventions. Educators need frameworks for assessing reflective contributions, balancing group dynamics, and nurturing psychological safety in both online and offline collaboration.
- Mental Well-Being and Reflective Practice: Future versions can integrate reflective prompts, emotion-aware feedback systems, and peer support networks within the platform to promote mental well-being alongside academic performance. Such features can transform collaboration into a more mindful, empathetic process.